

# BFS Traversal

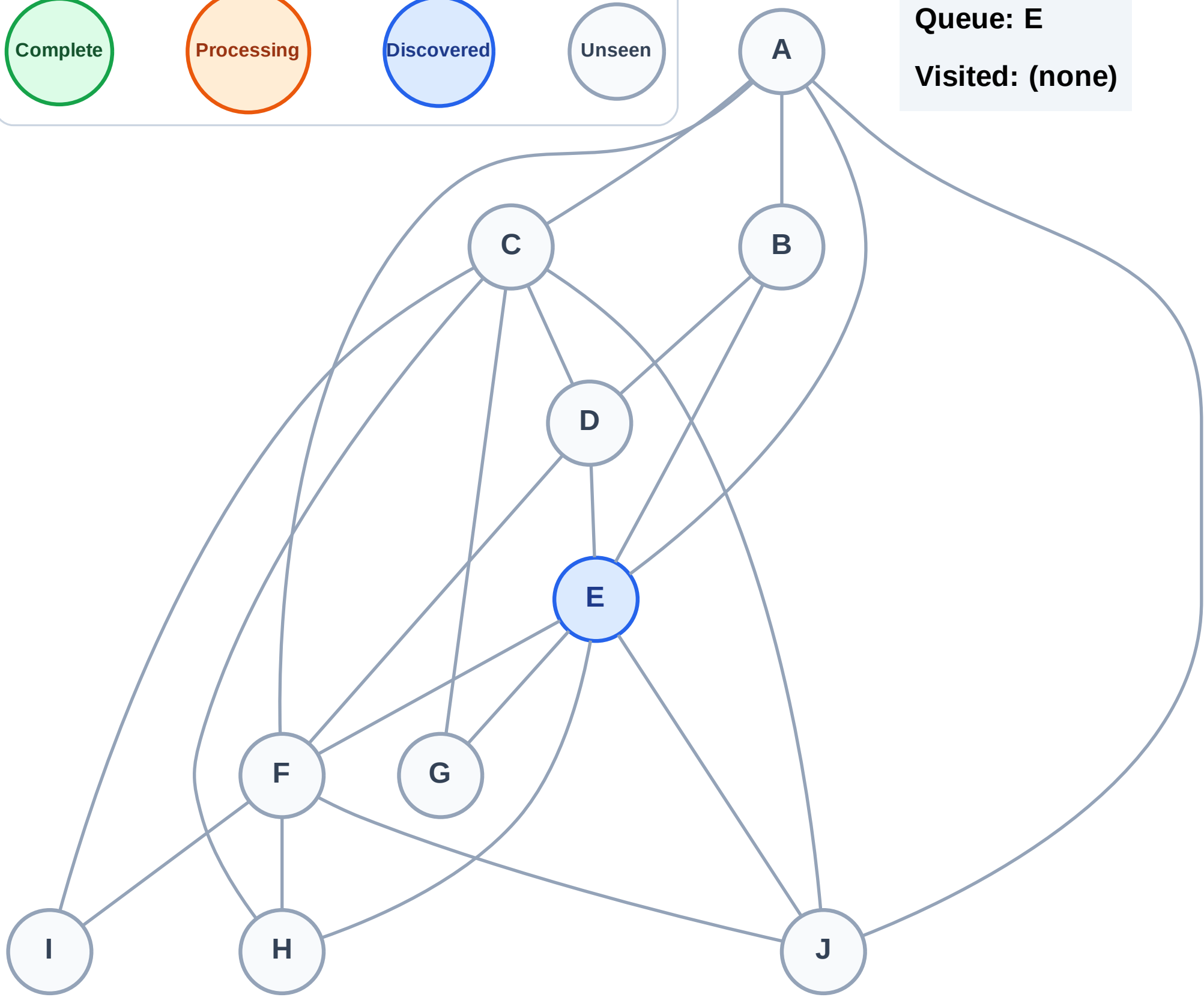
Step 1: Start at E

## Legend



Queue: E

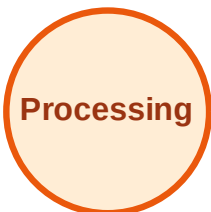
Visited: (none)



# BFS Traversal

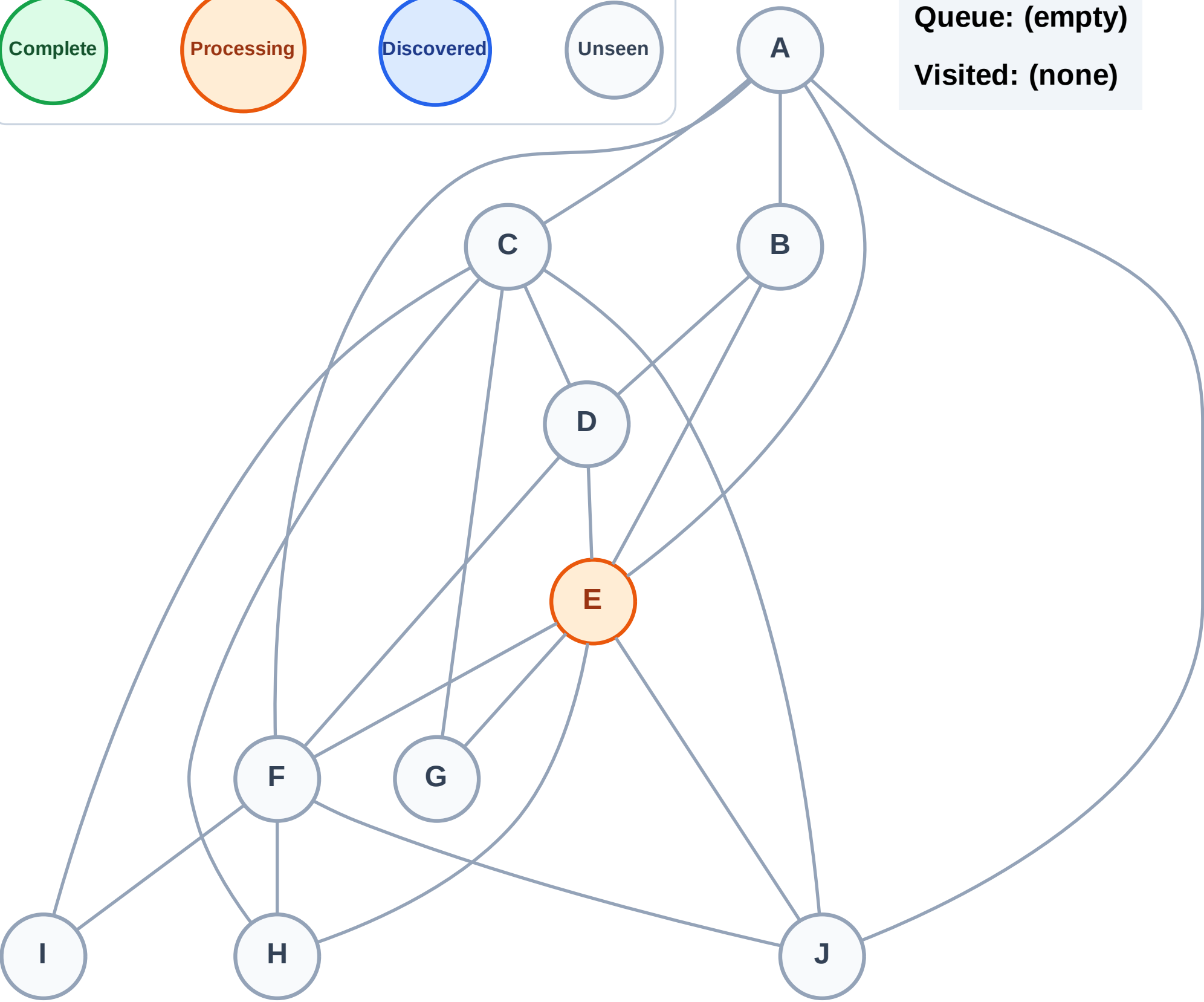
Step 2: Process node E

## Legend



Queue: (empty)

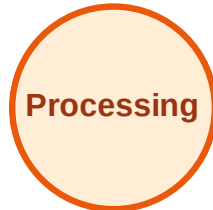
Visited: (none)



# BFS Traversal

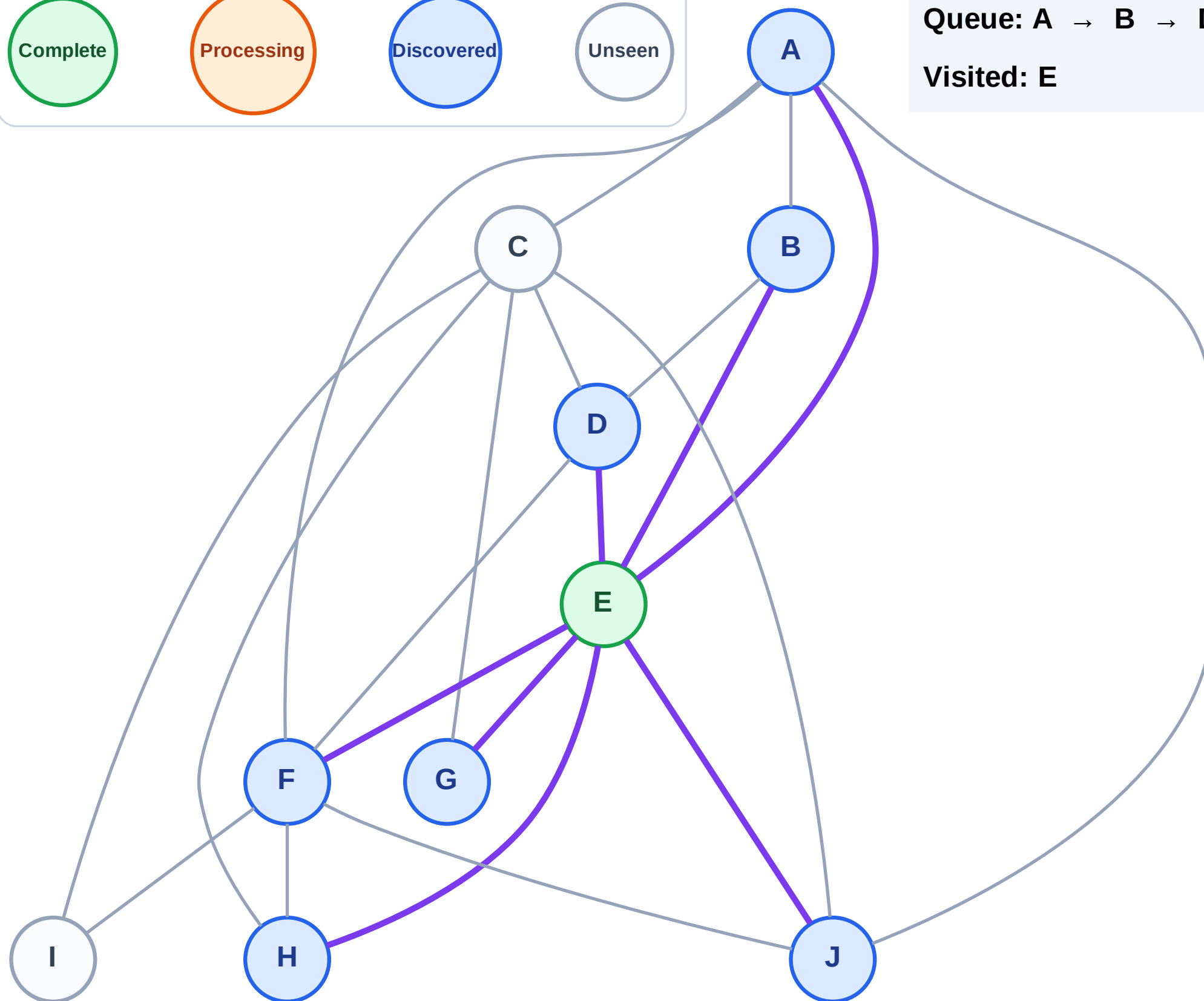
Step 3: Discover A, B, D, F, G, H, J from E

## Legend



Queue: A → B → D → F → G → H → J

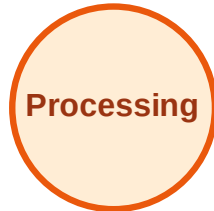
Visited: E



# BFS Traversal

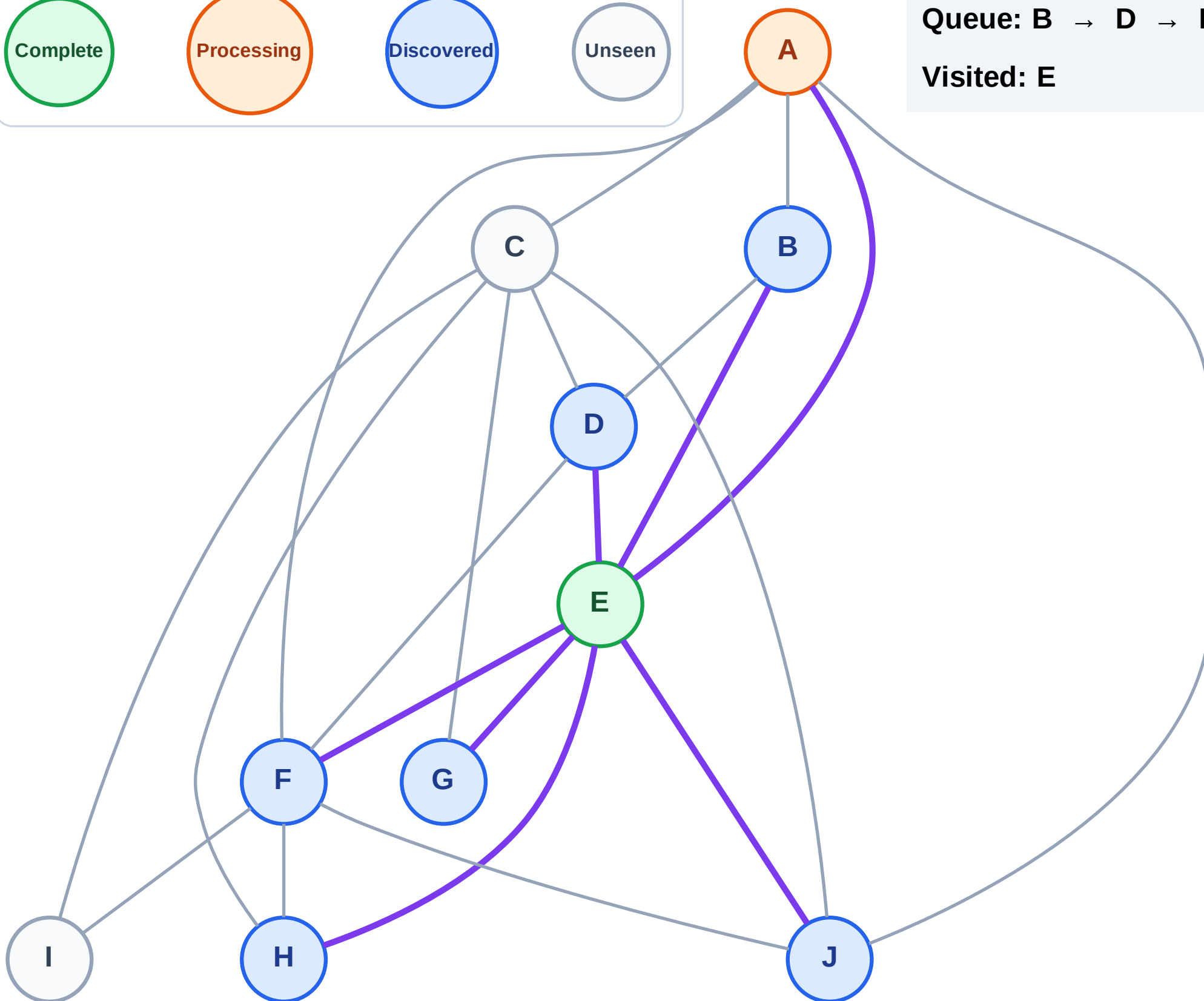
Step 4: Process node A

## Legend



Queue: B → D → F → G → H → J

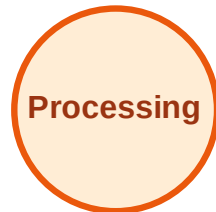
Visited: E



# BFS Traversal

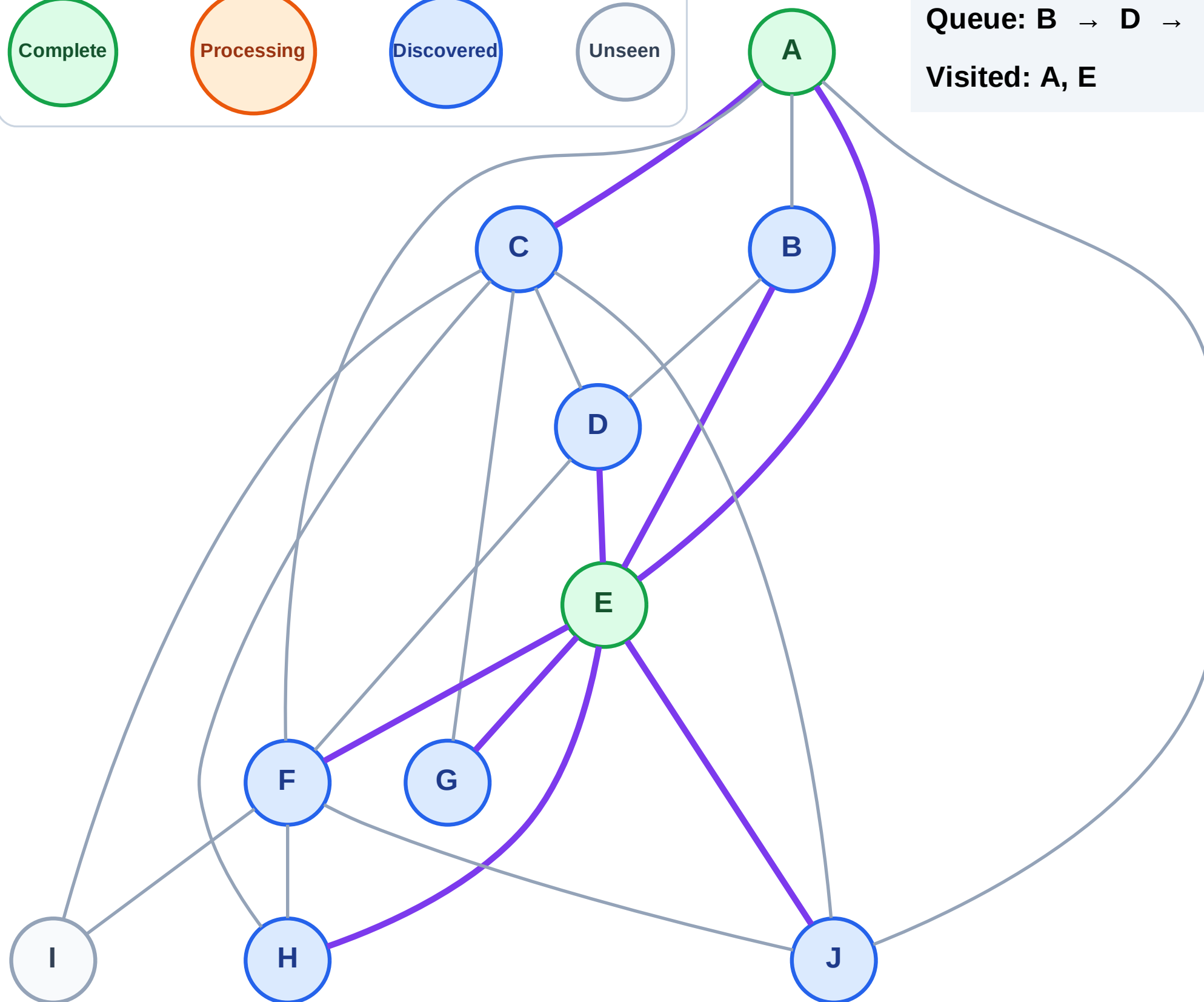
Step 5: Discover C from A

## Legend



Queue: B → D → F → G → H → J → C

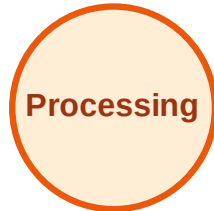
Visited: A, E



# BFS Traversal

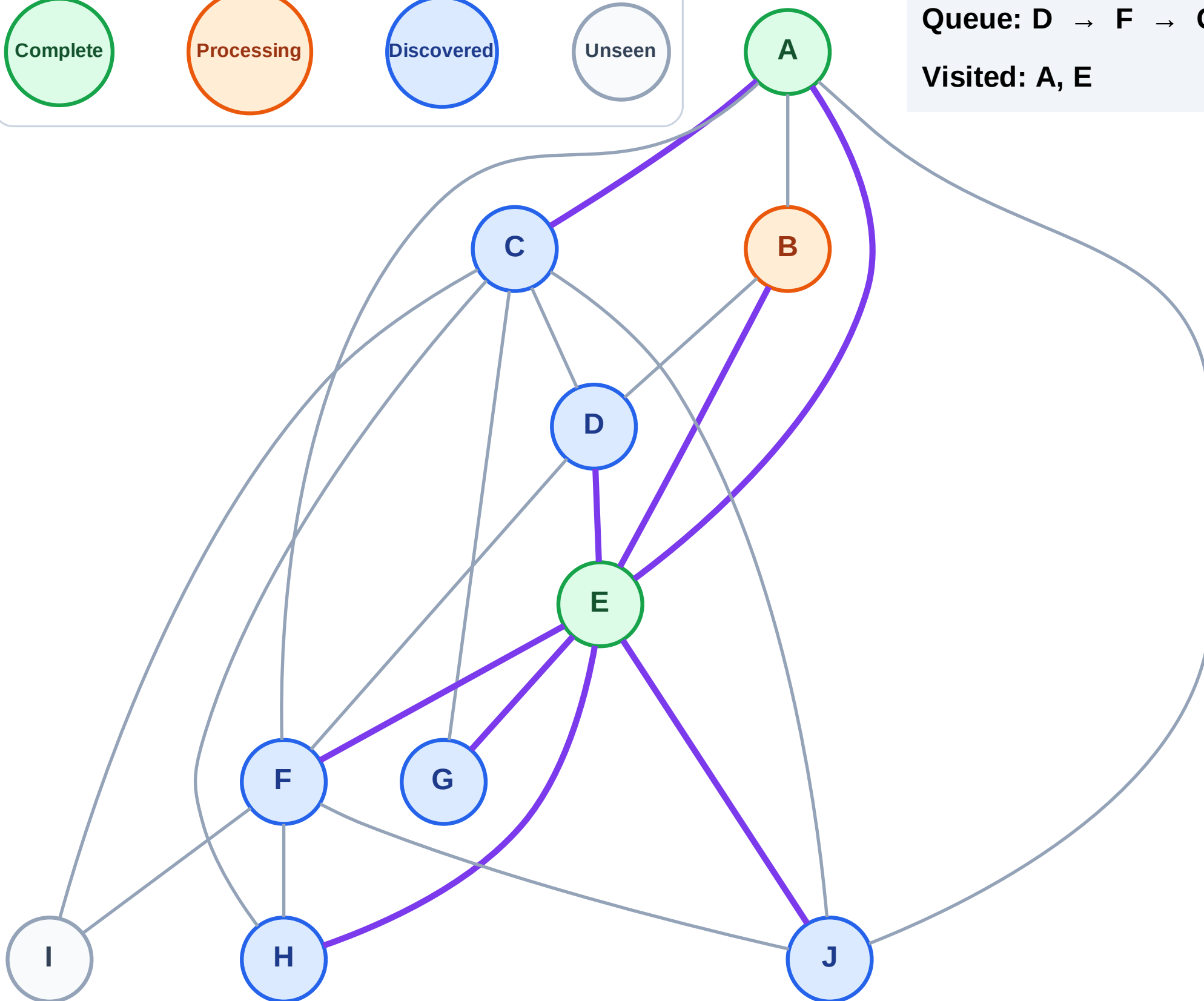
Step 6: Process node B

## Legend



Queue: D → F → G → H → J → C

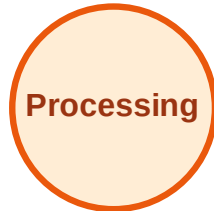
Visited: A, E



# BFS Traversal

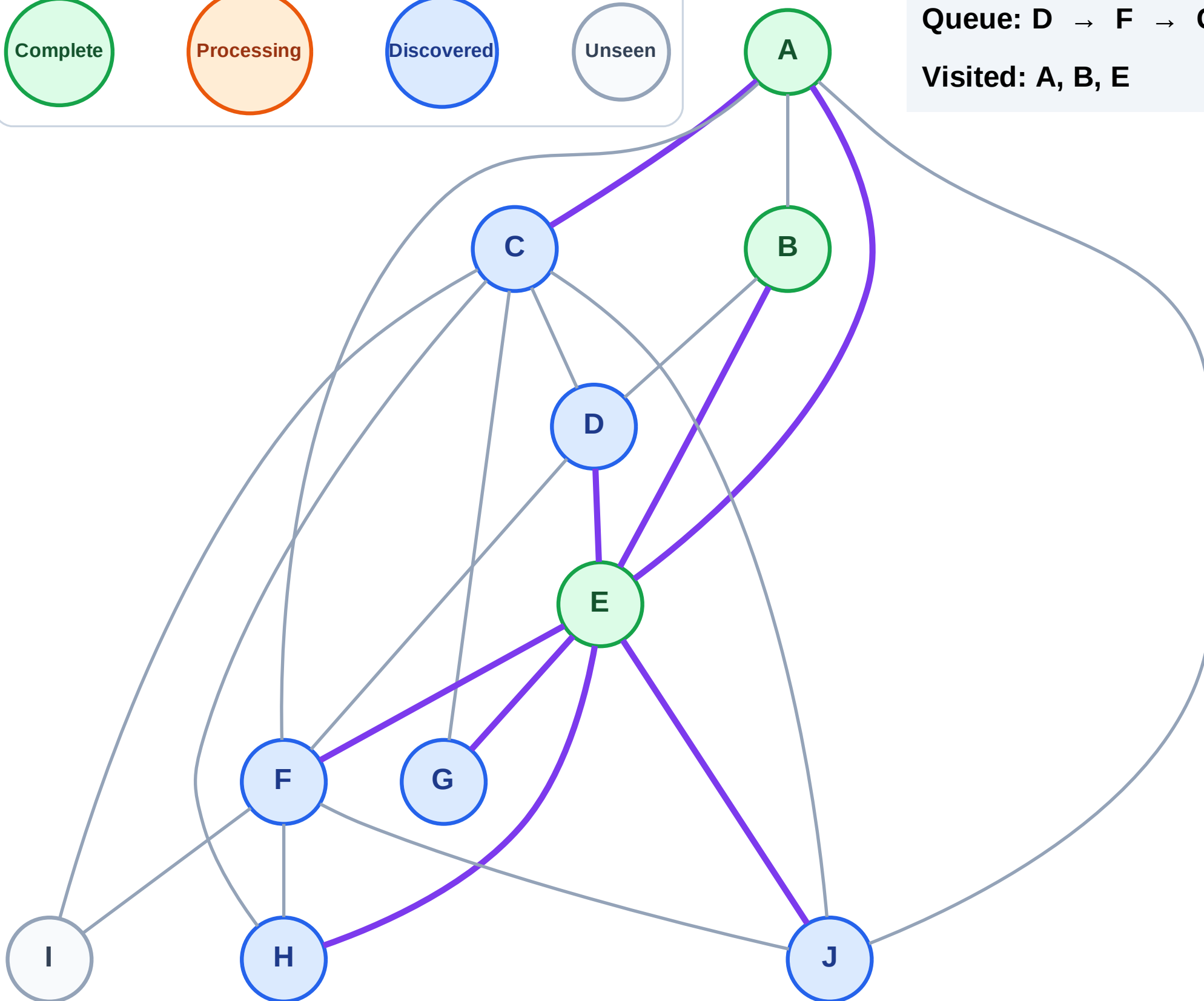
Step 7: No new neighbors from B

## Legend



Queue: D → F → G → H → J → C

Visited: A, B, E



# BFS Traversal

Step 8: Process node D

Legend

Complete

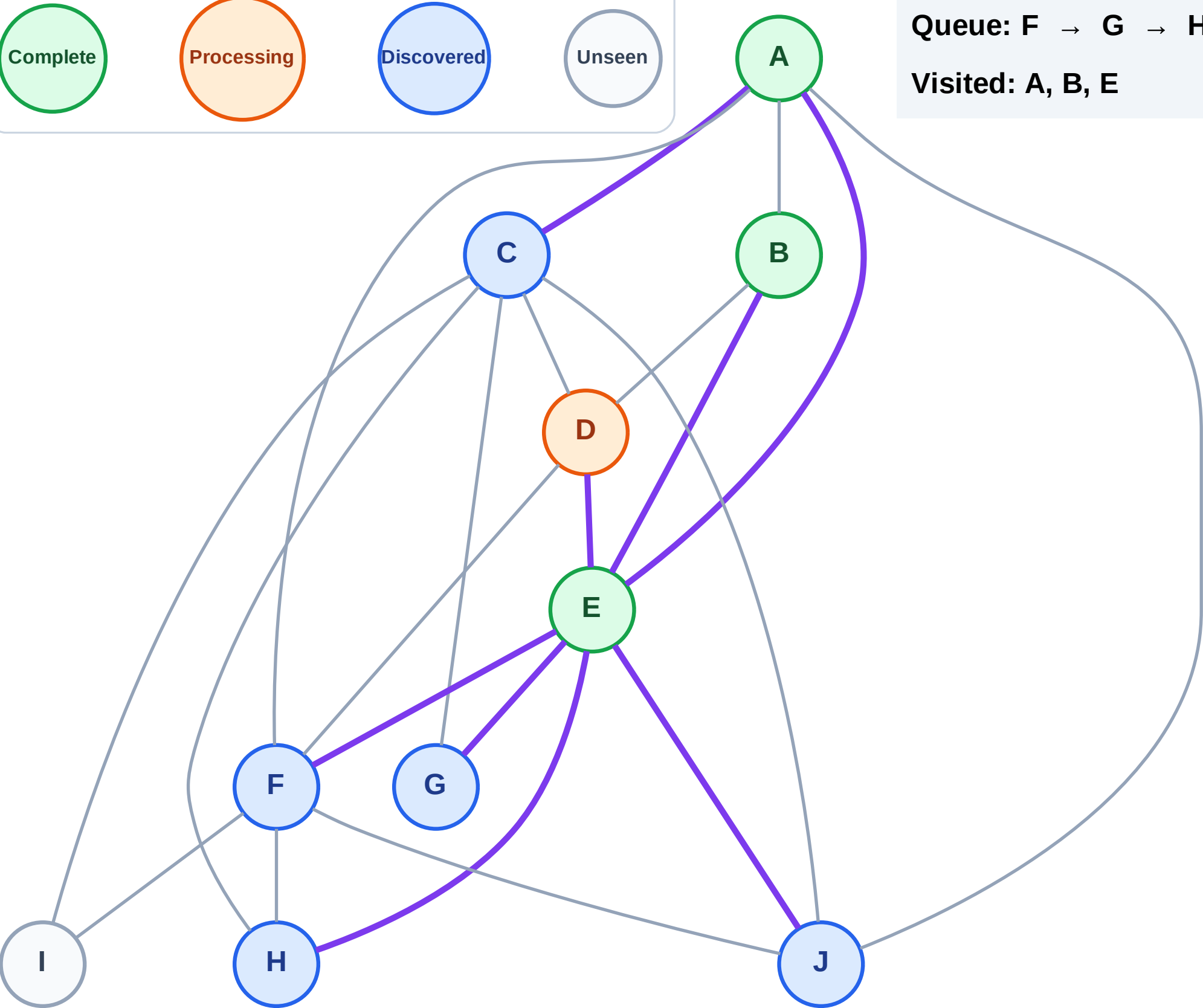
Processing

Discovered

Unseen

Queue: F → G → H → J → C

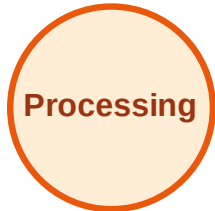
Visited: A, B, E



# BFS Traversal

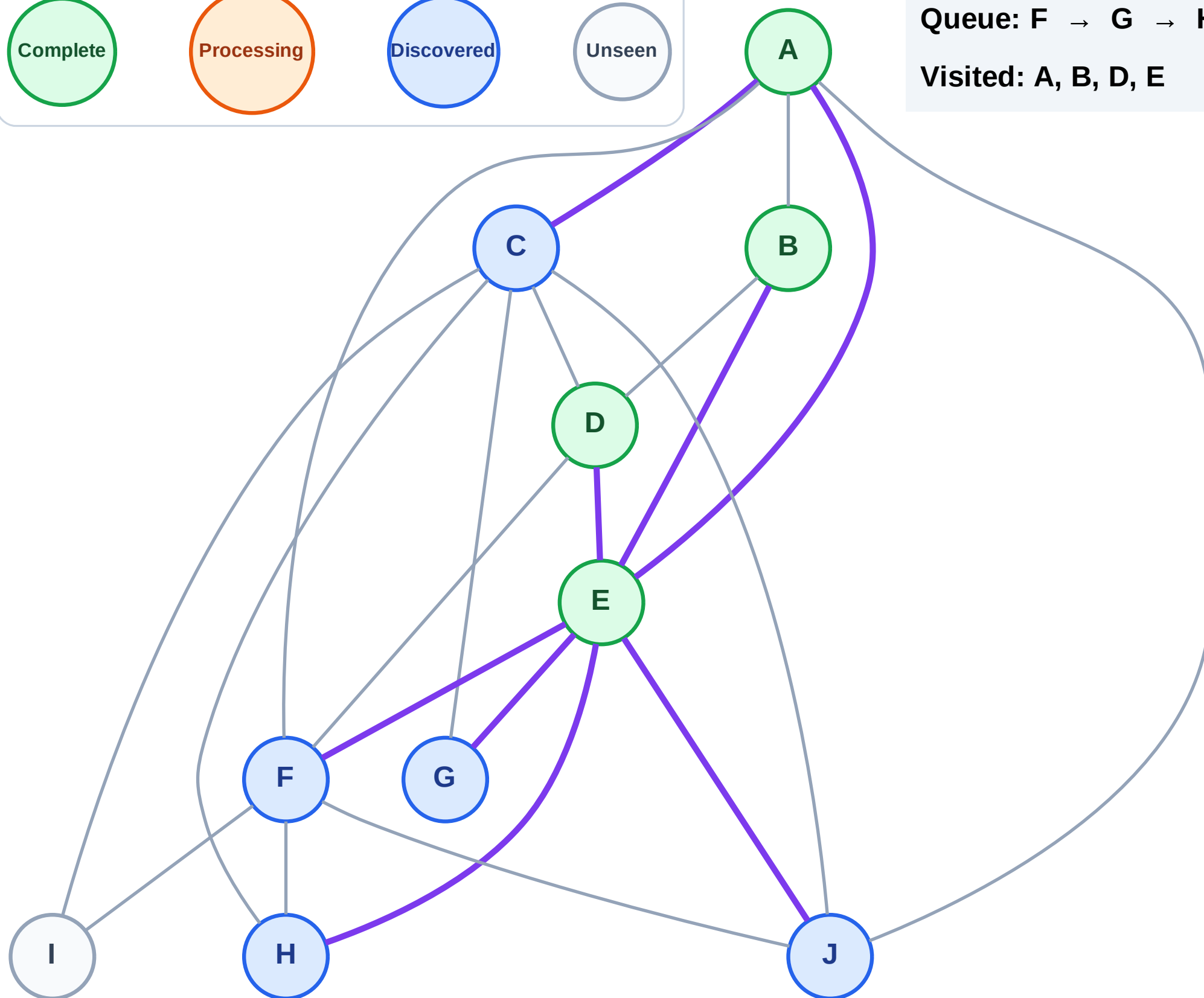
Step 9: No new neighbors from D

## Legend



Queue: F → G → H → J → C

Visited: A, B, D, E



# BFS Traversal

Step 10: Process node F

Legend

Complete

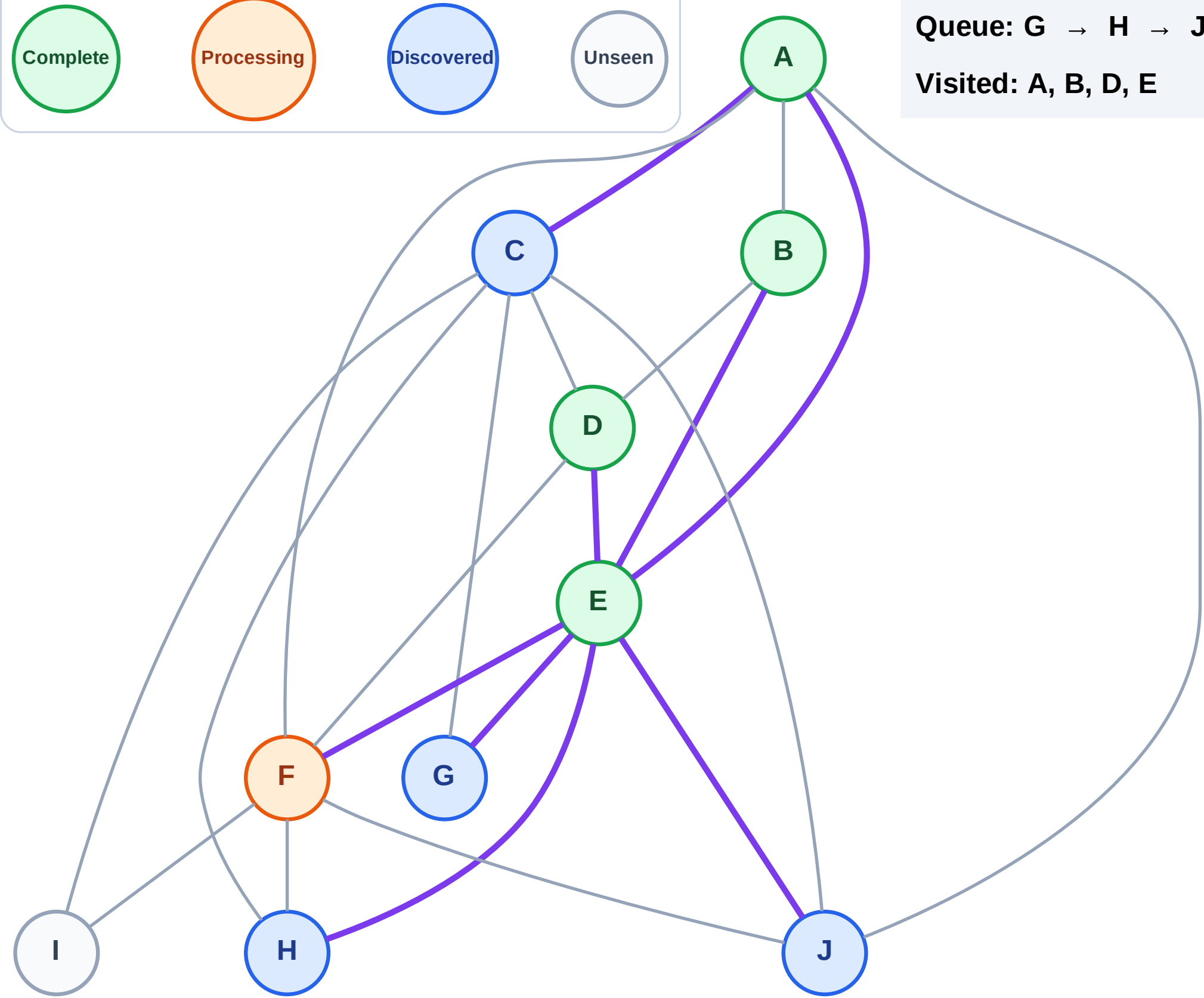
Processing

Discovered

Unseen

Queue: G → H → J → C

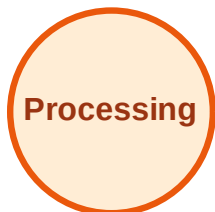
Visited: A, B, D, E



# BFS Traversal

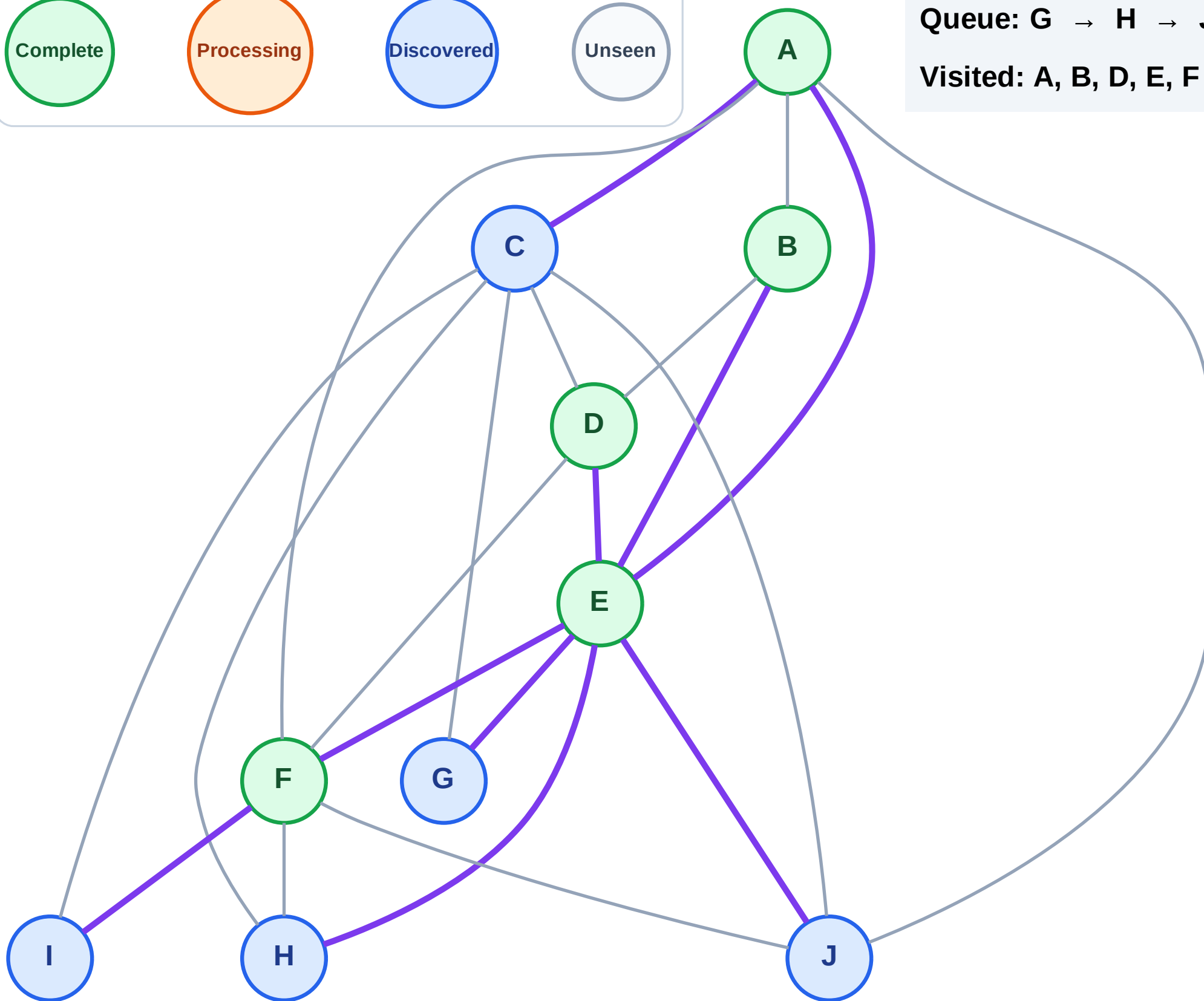
Step 11: Discover I from F

## Legend



Queue: G → H → J → C → I

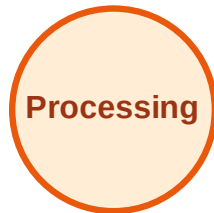
Visited: A, B, D, E, F



# BFS Traversal

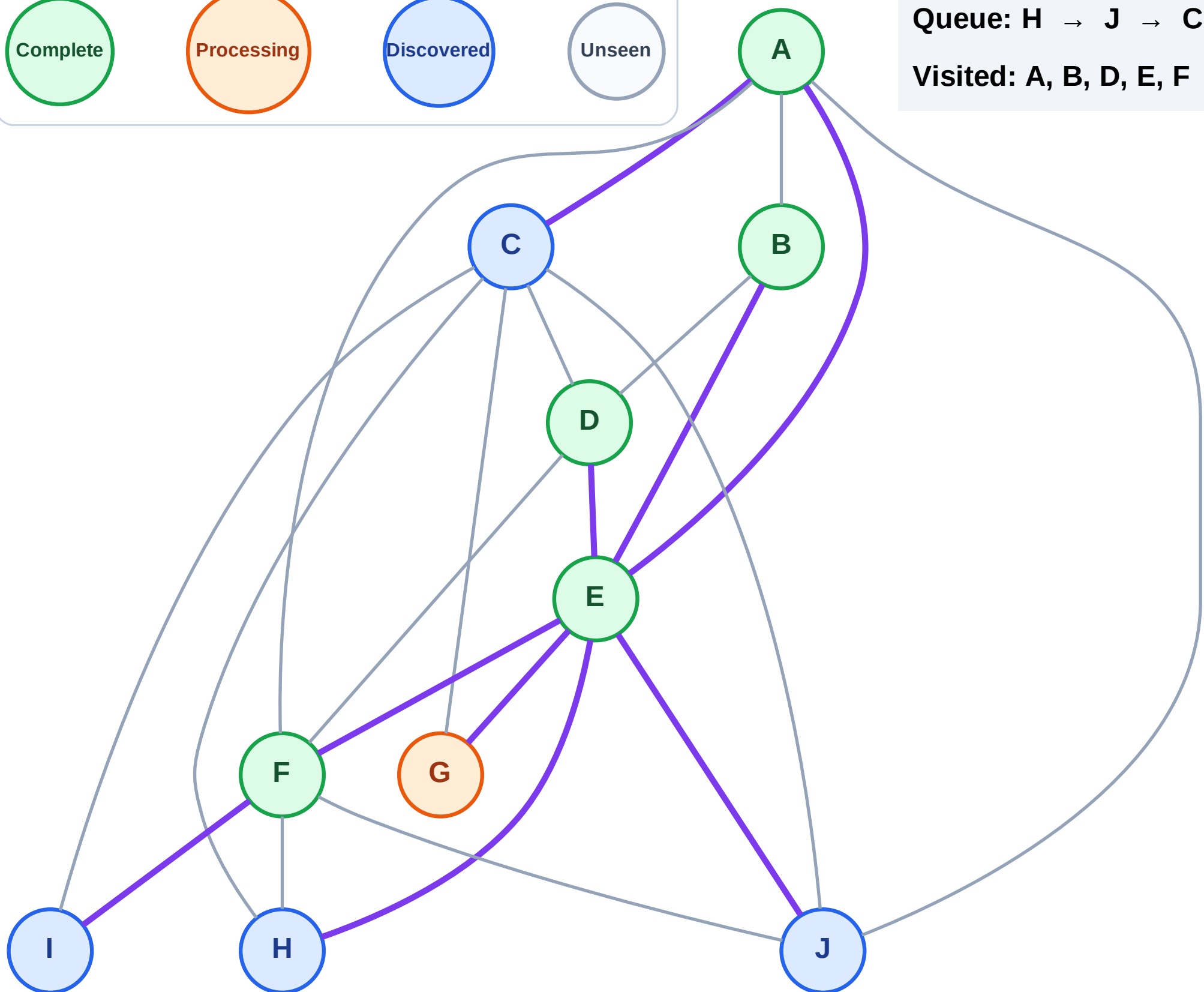
Step 12: Process node G

## Legend



Queue: H → J → C → I

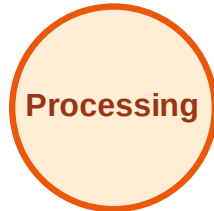
Visited: A, B, D, E, F



# BFS Traversal

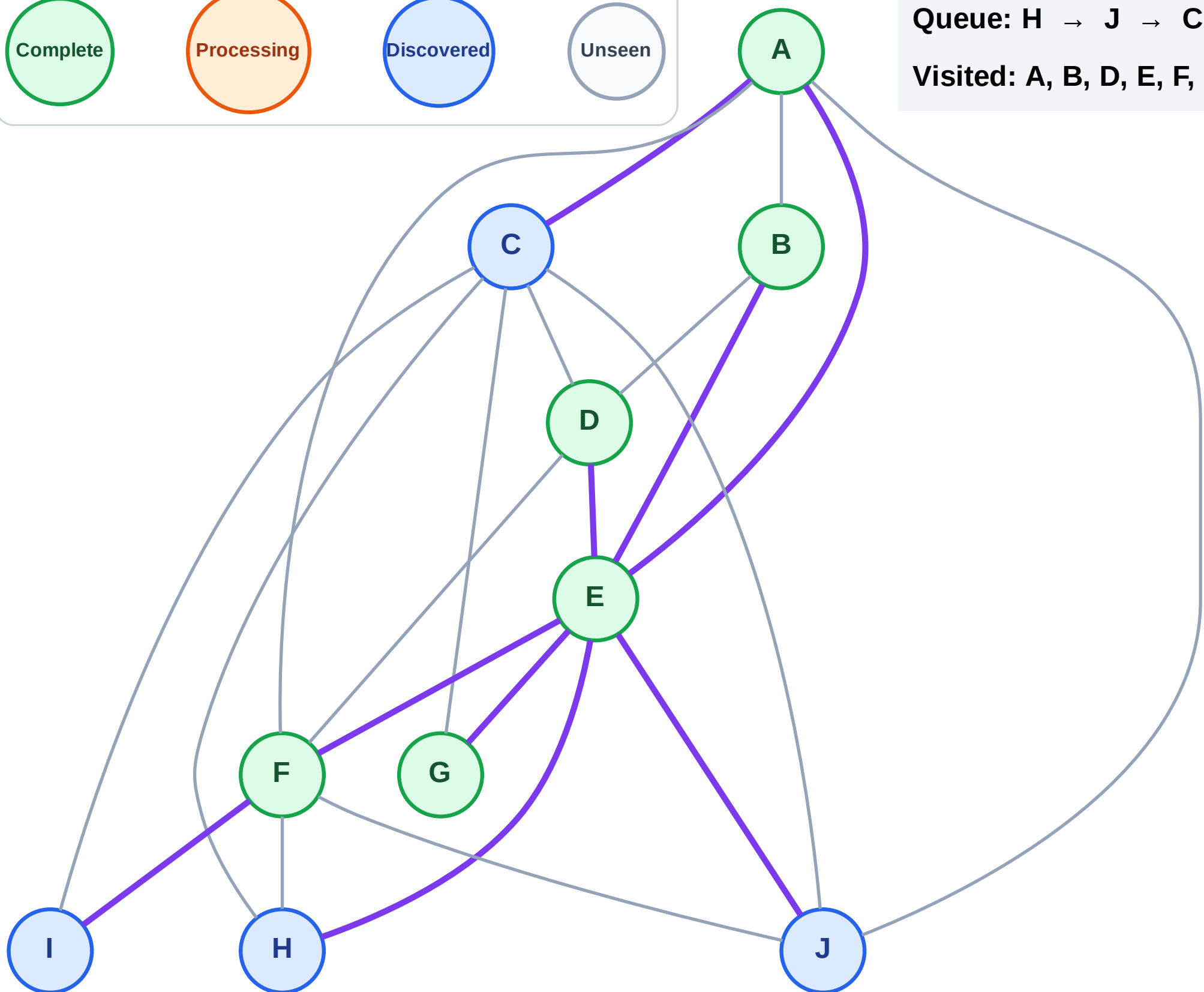
Step 13: No new neighbors from G

## Legend



Queue: H → J → C → I

Visited: A, B, D, E, F, G



# BFS Traversal

Step 14: Process node H

Legend

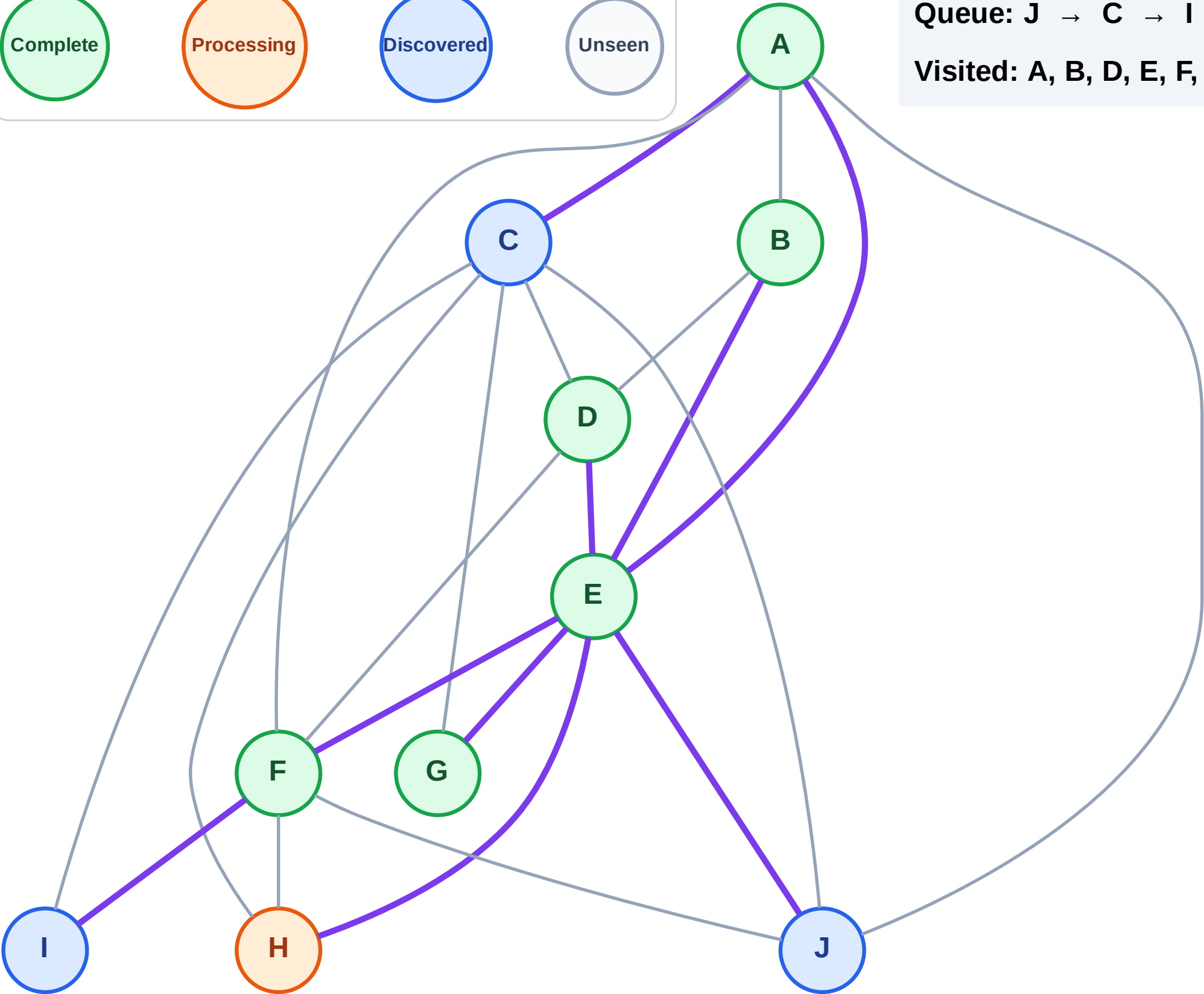
Complete

Processing

Discovered

Unseen

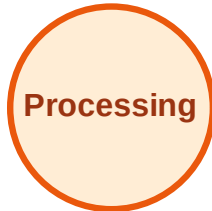
Queue: J → C → I  
Visited: A, B, D, E, F, G



# BFS Traversal

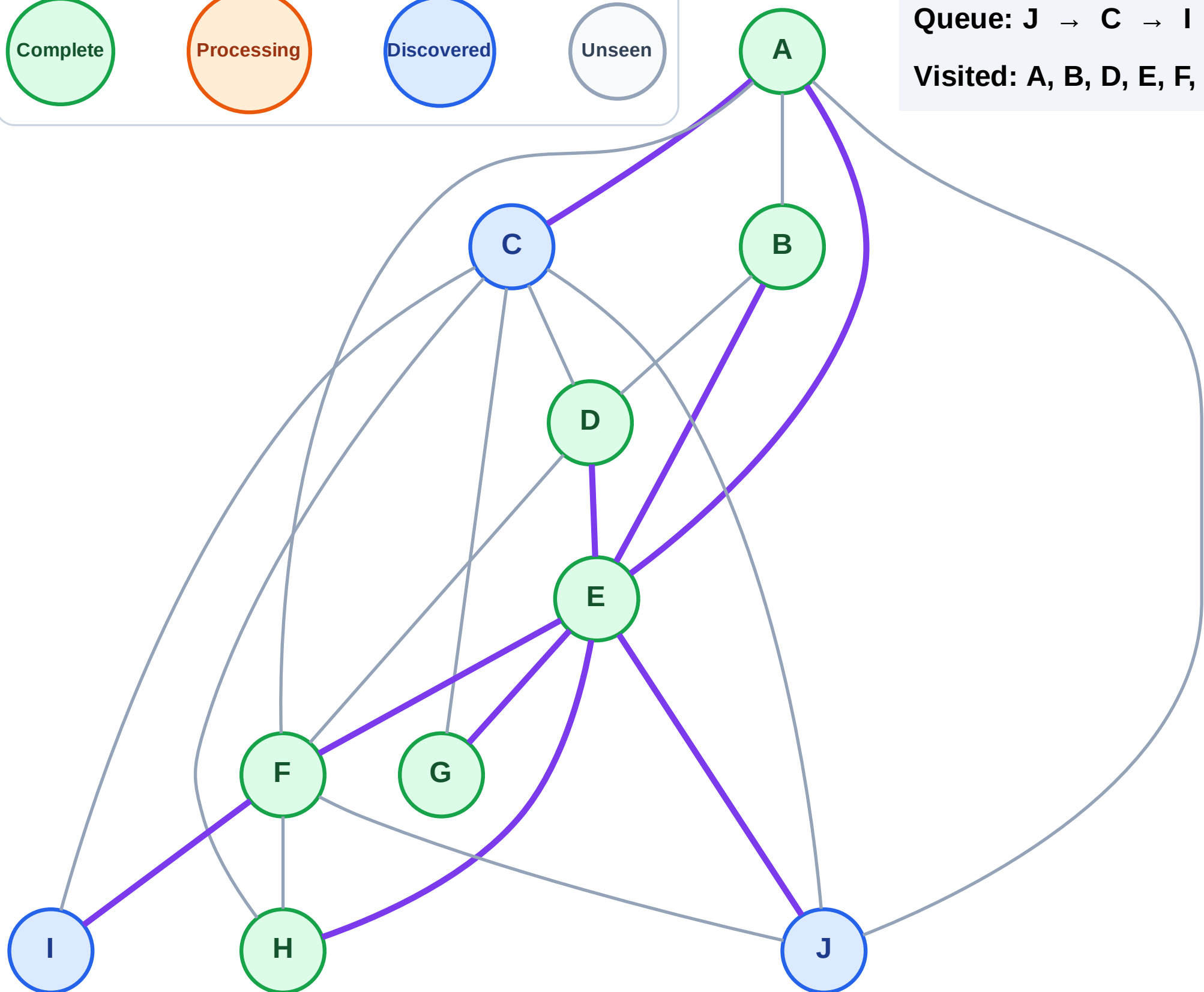
Step 15: No new neighbors from H

## Legend



Queue: J → C → I

Visited: A, B, D, E, F, G, H



# BFS Traversal

Step 16: Process node J

Legend

Complete

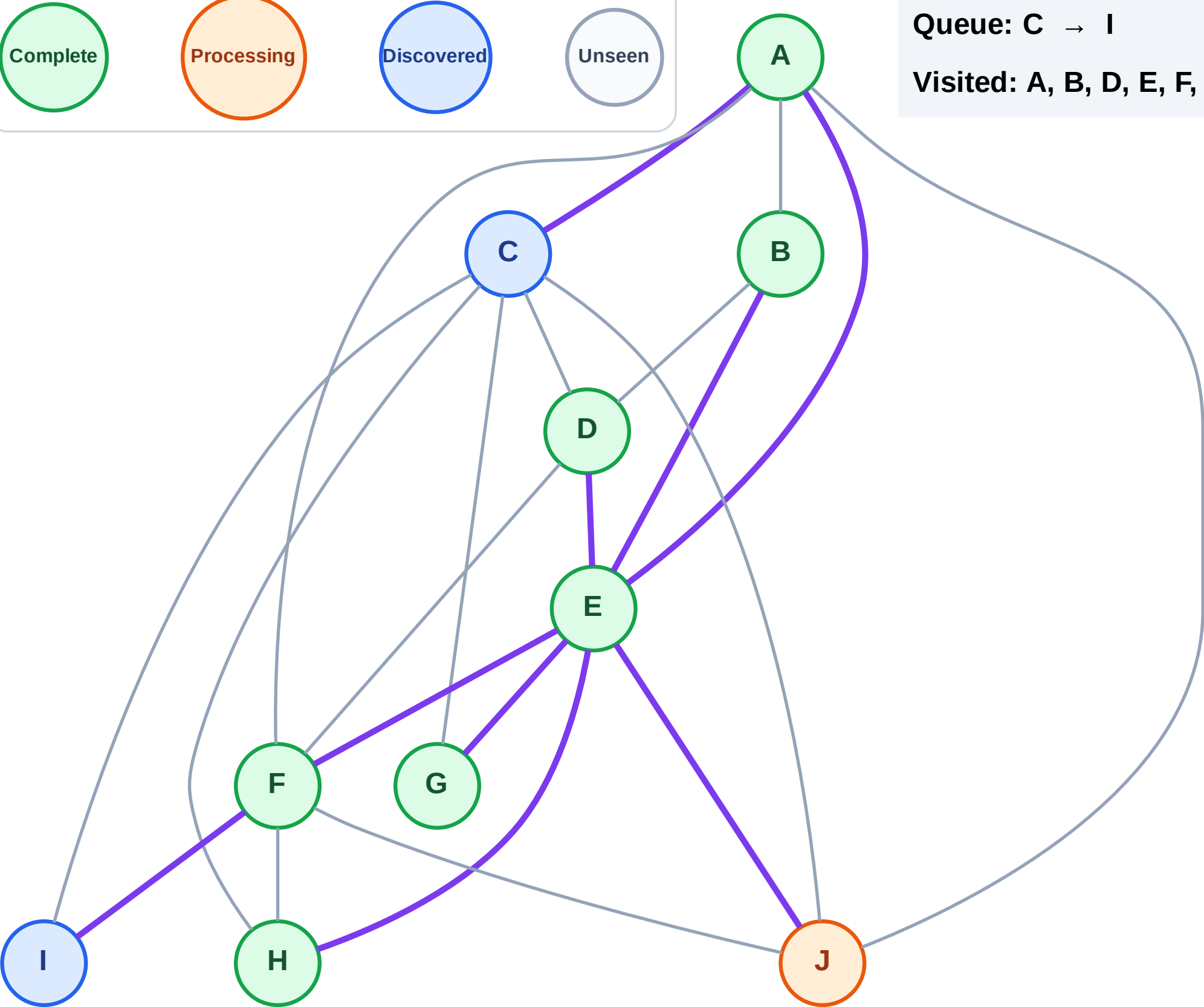
Processing

Discovered

Unseen

Queue: C → I

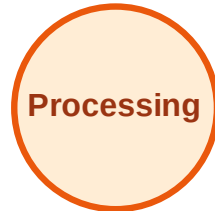
Visited: A, B, D, E, F, G, H



# BFS Traversal

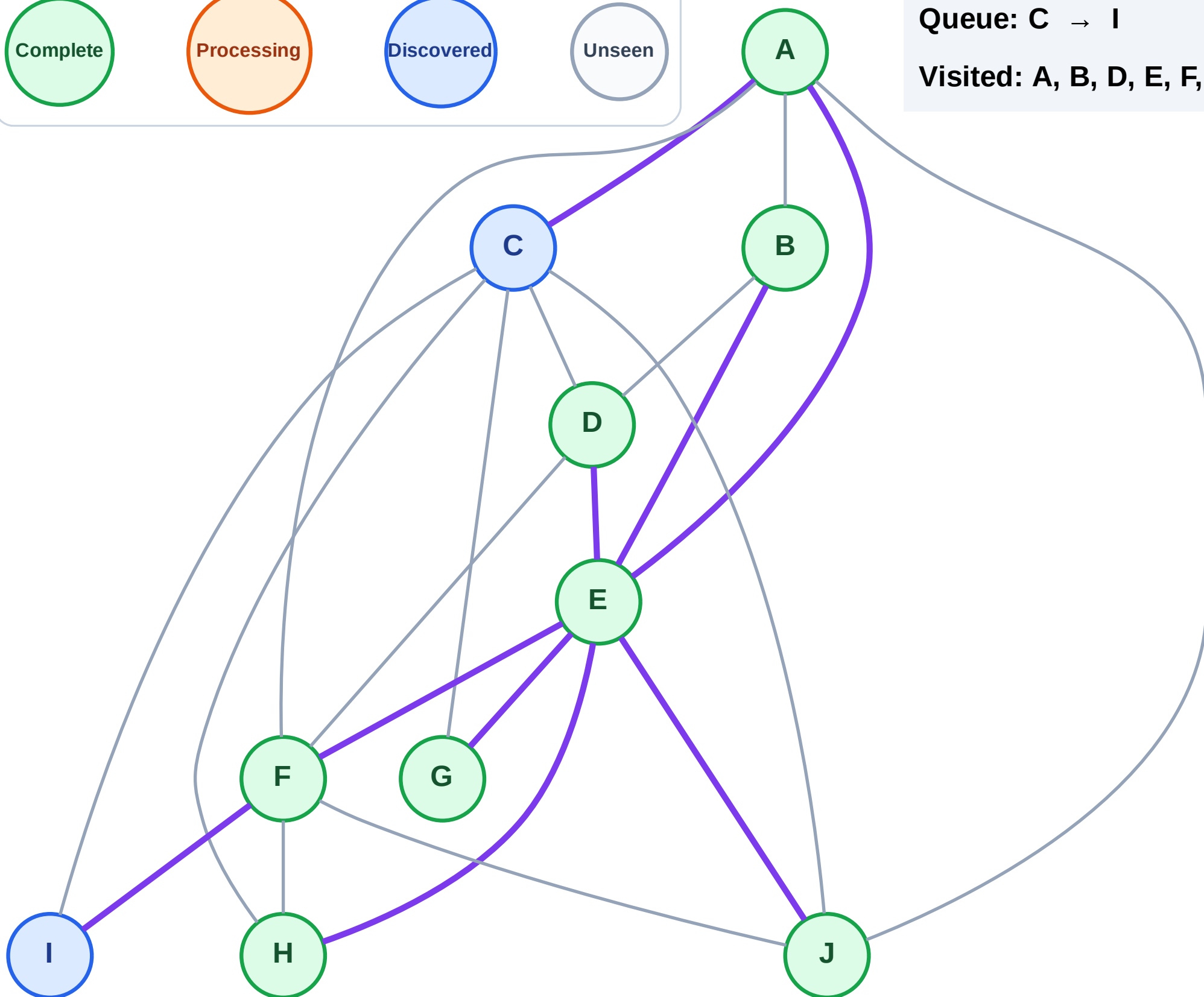
Step 17: No new neighbors from J

## Legend



Queue: C → I

Visited: A, B, D, E, F, G, H, J



# BFS Traversal

Step 18: Process node C

Legend

Complete

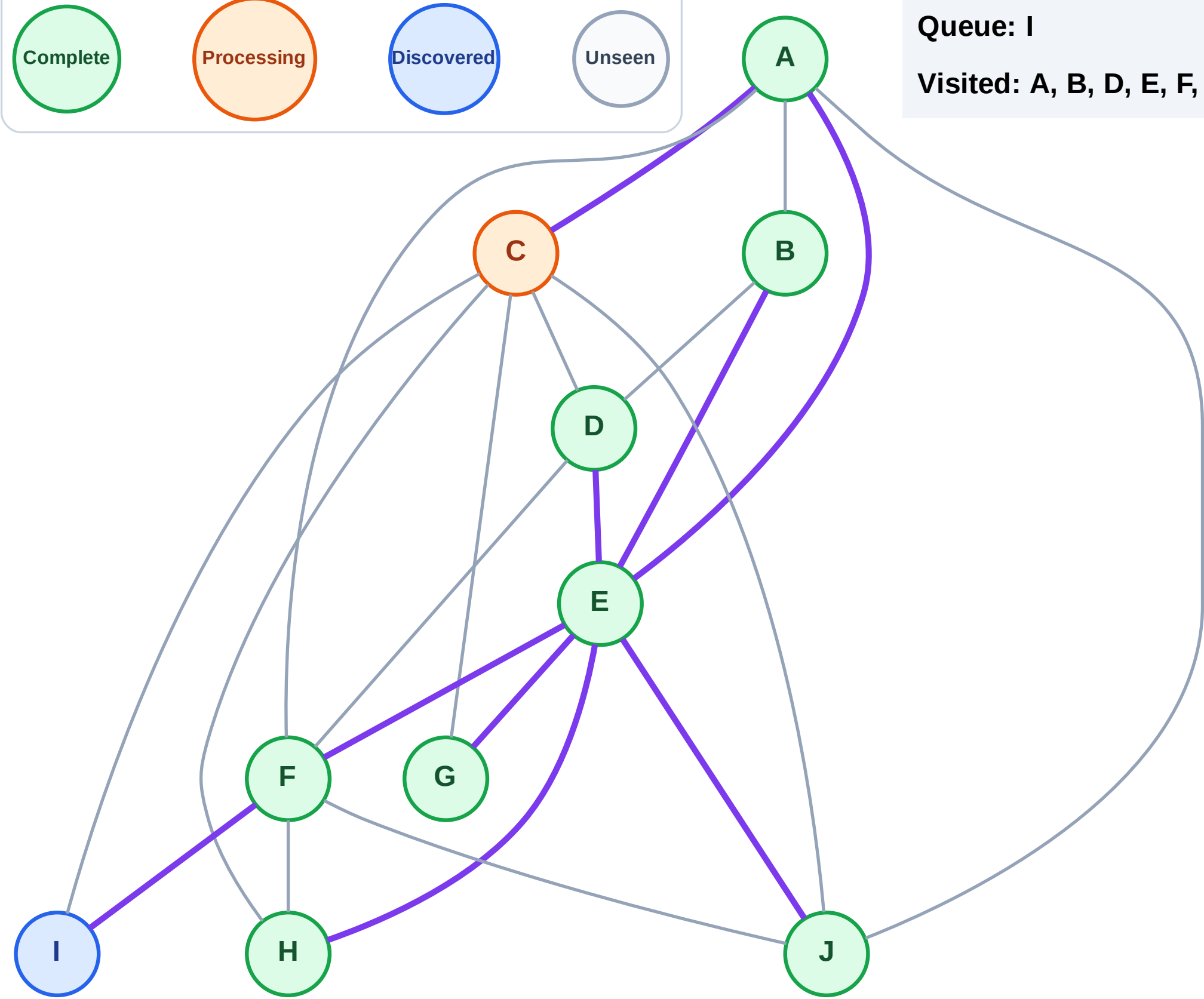
Processing

Discovered

Unseen

Queue: I

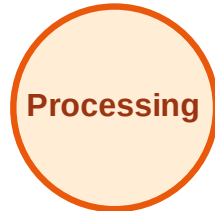
Visited: A, B, D, E, F, G, H, J



# BFS Traversal

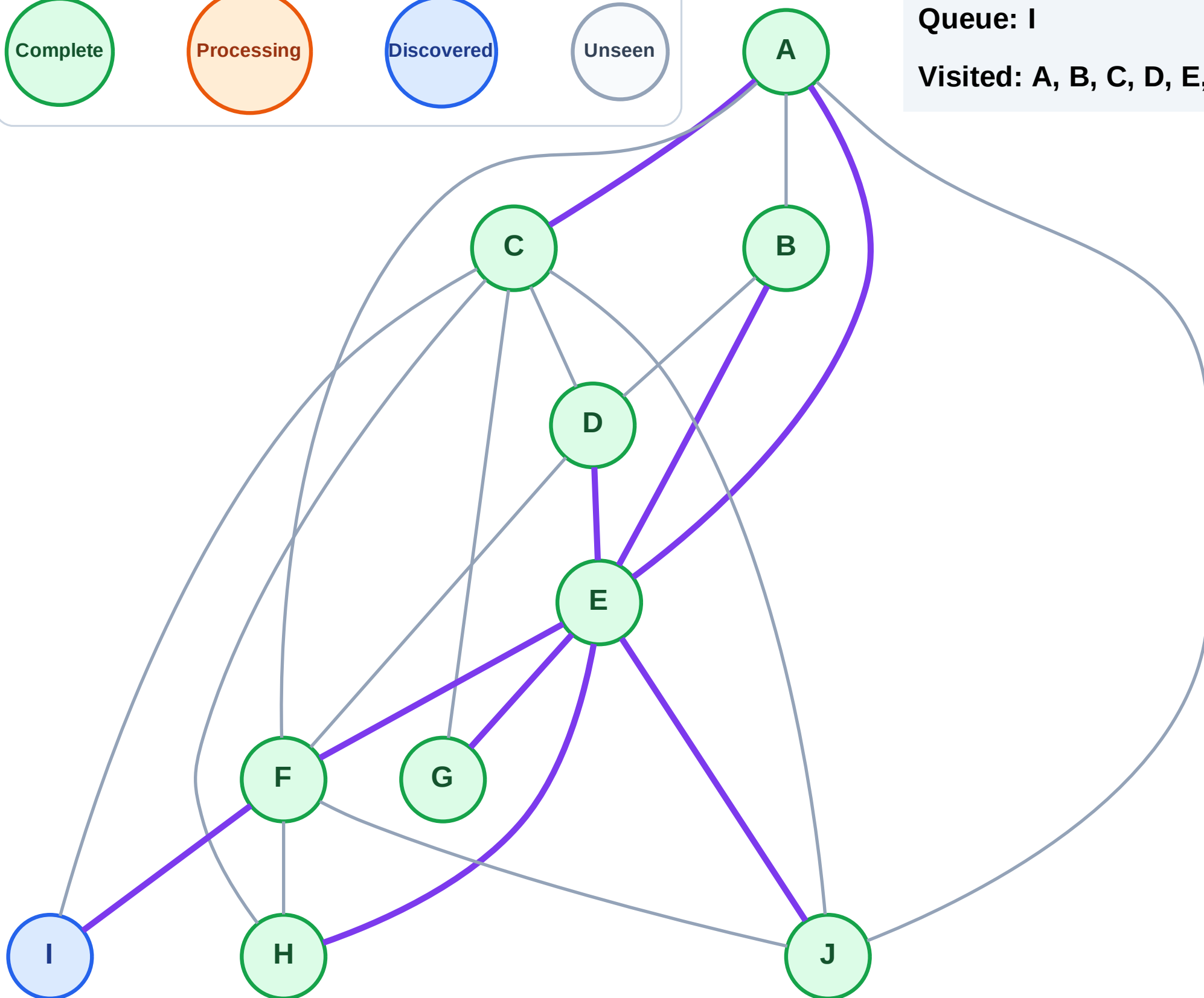
Step 19: No new neighbors from C

## Legend



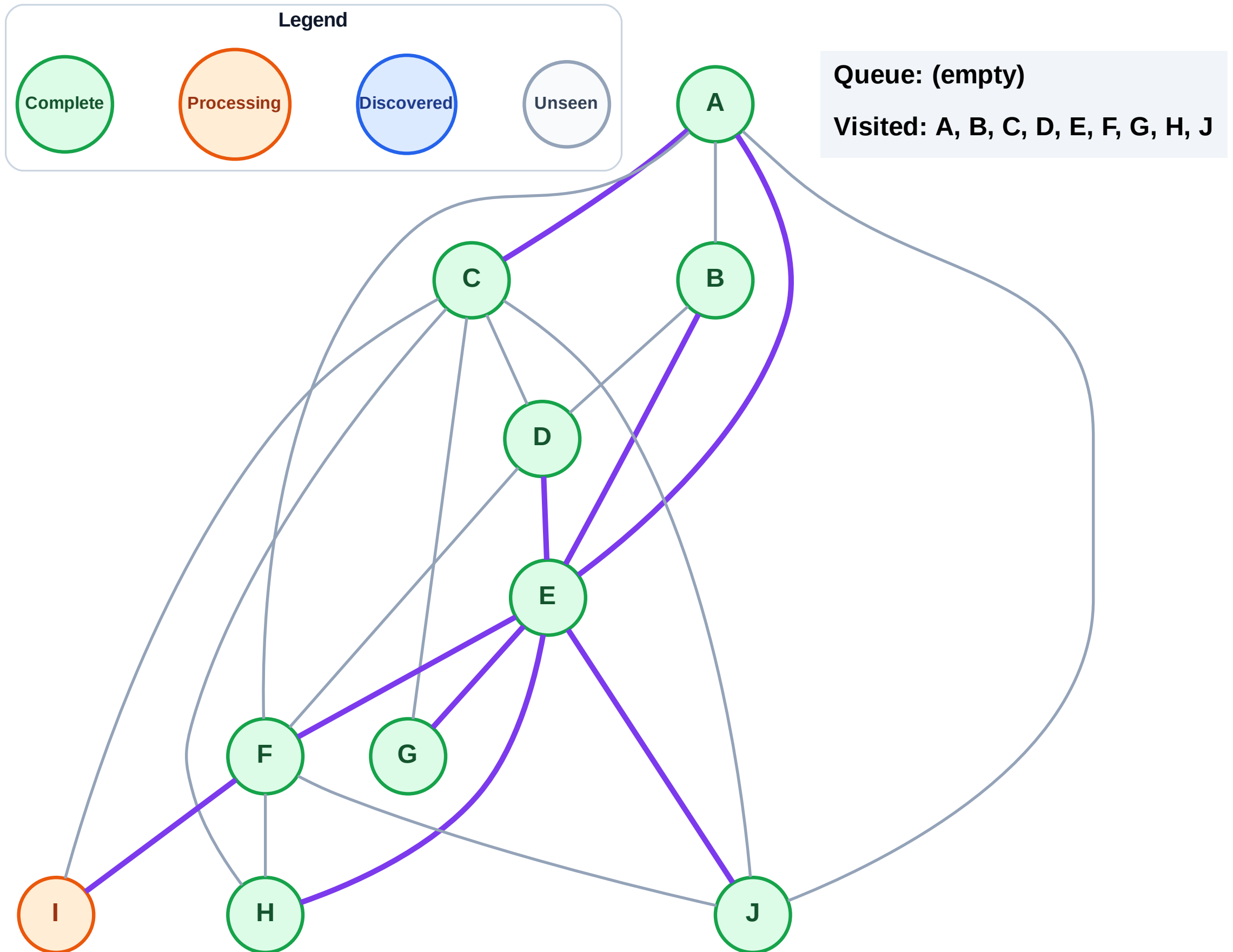
Queue: I

Visited: A, B, C, D, E, F, G, H, J



## Step 20: Process node I

## Step 20: Process node I



# BFS Traversal

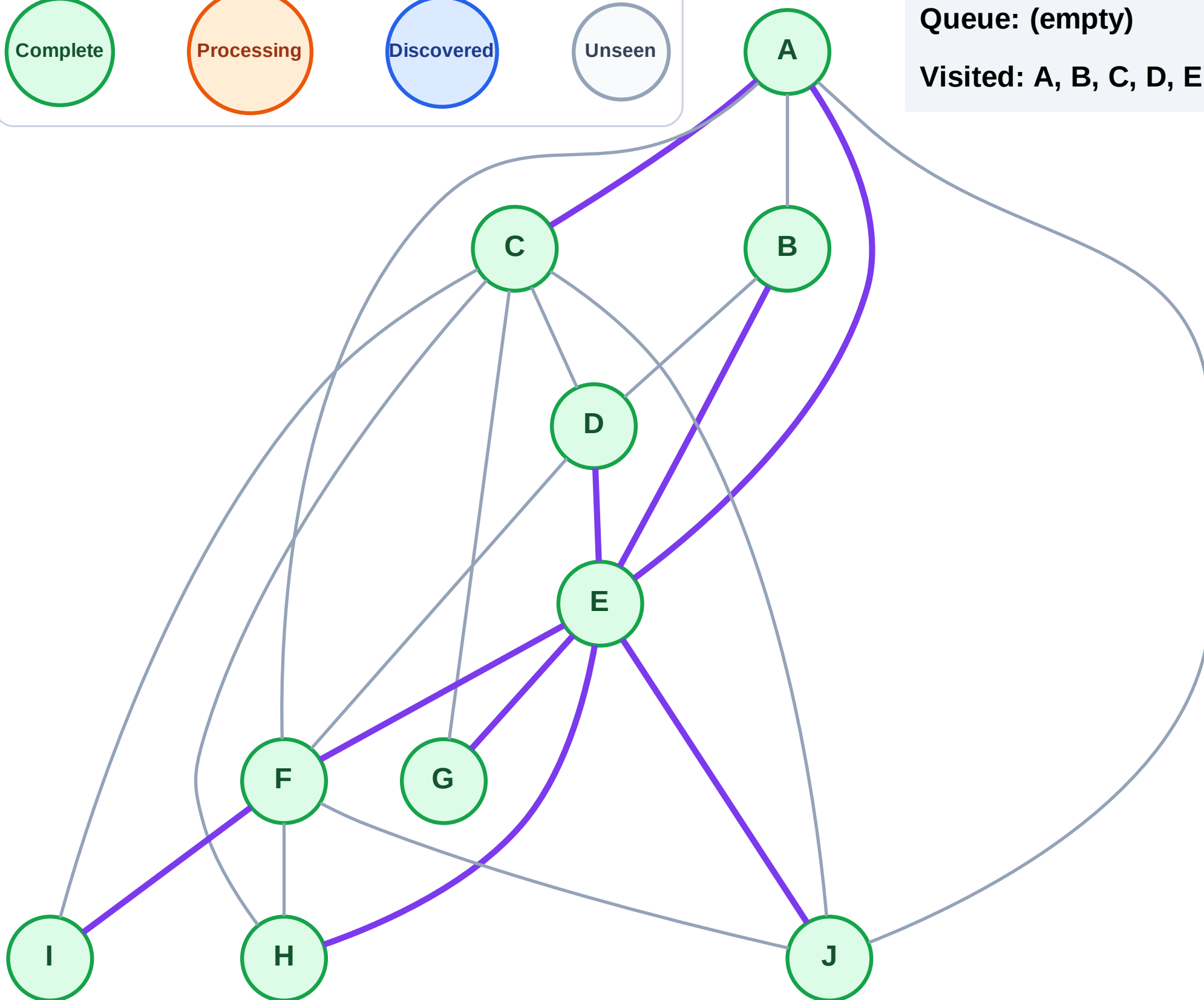
Step 21: No new neighbors from I

## Legend



Queue: (empty)

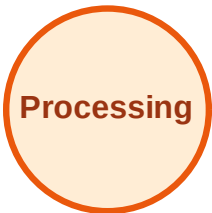
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

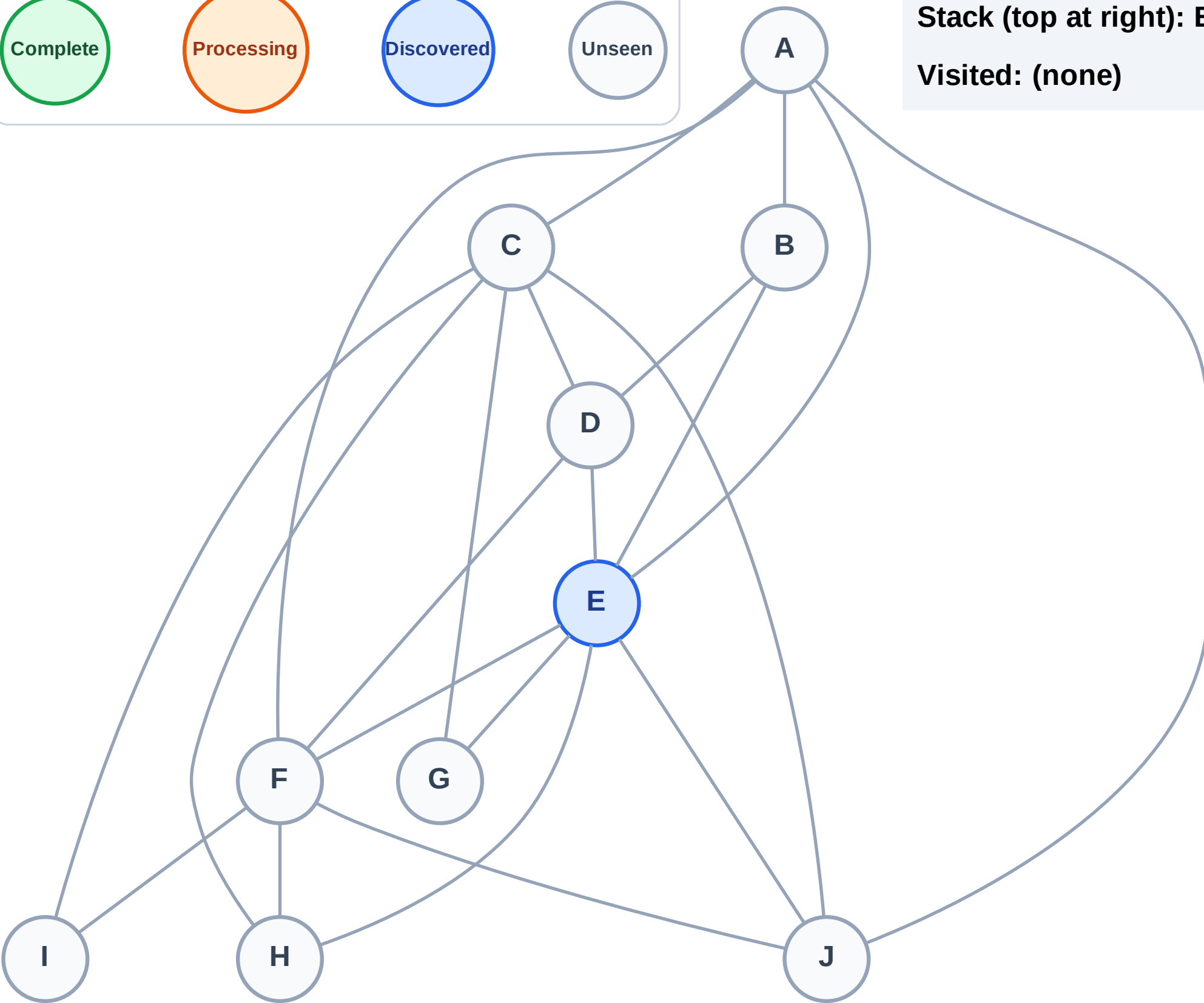
Step 1: Start at E

## Legend



Stack (top at right): E

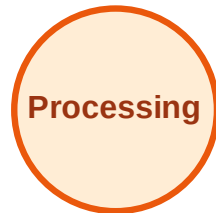
Visited: (none)



# DFS Traversal

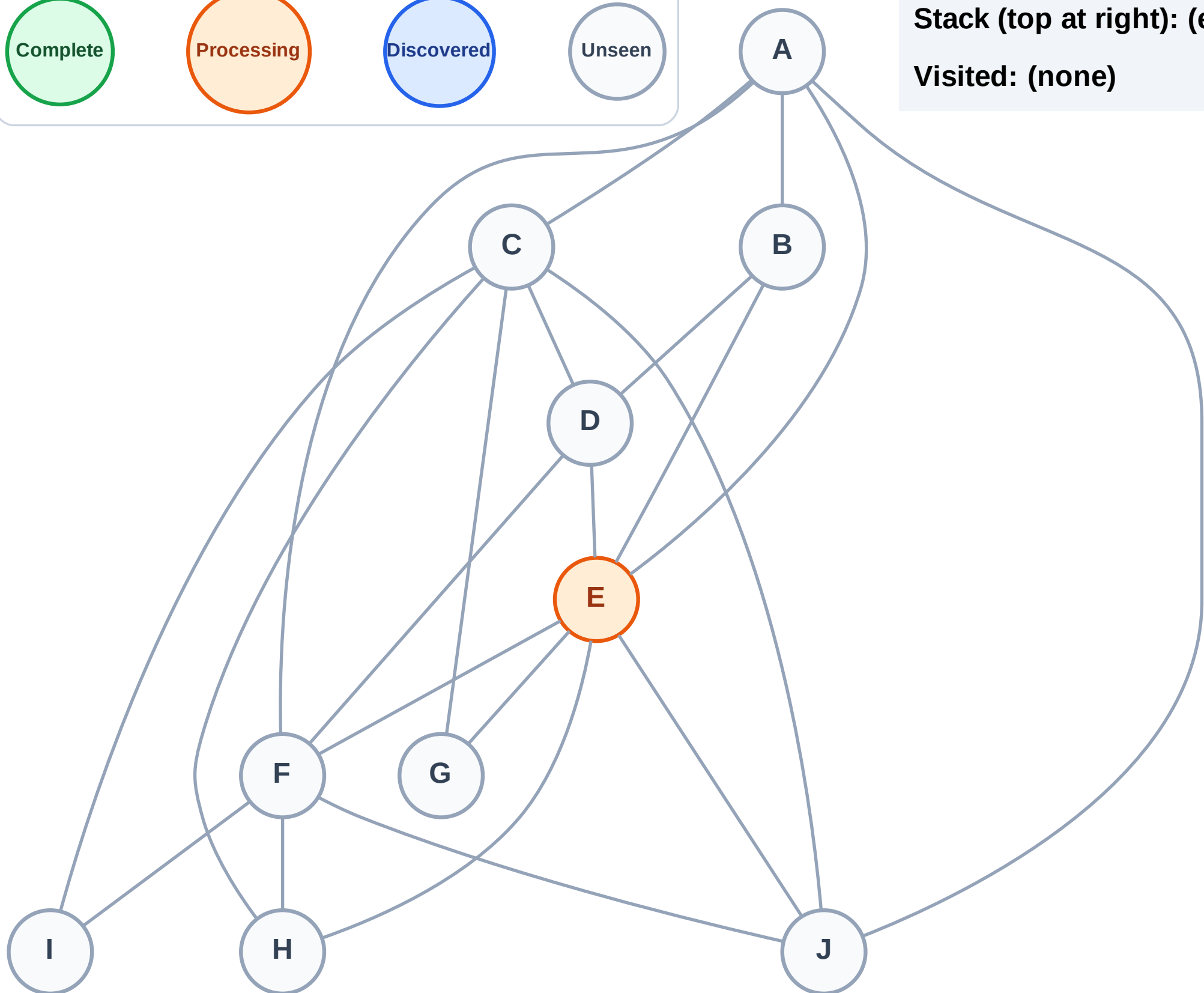
Step 2: Process node E

## Legend



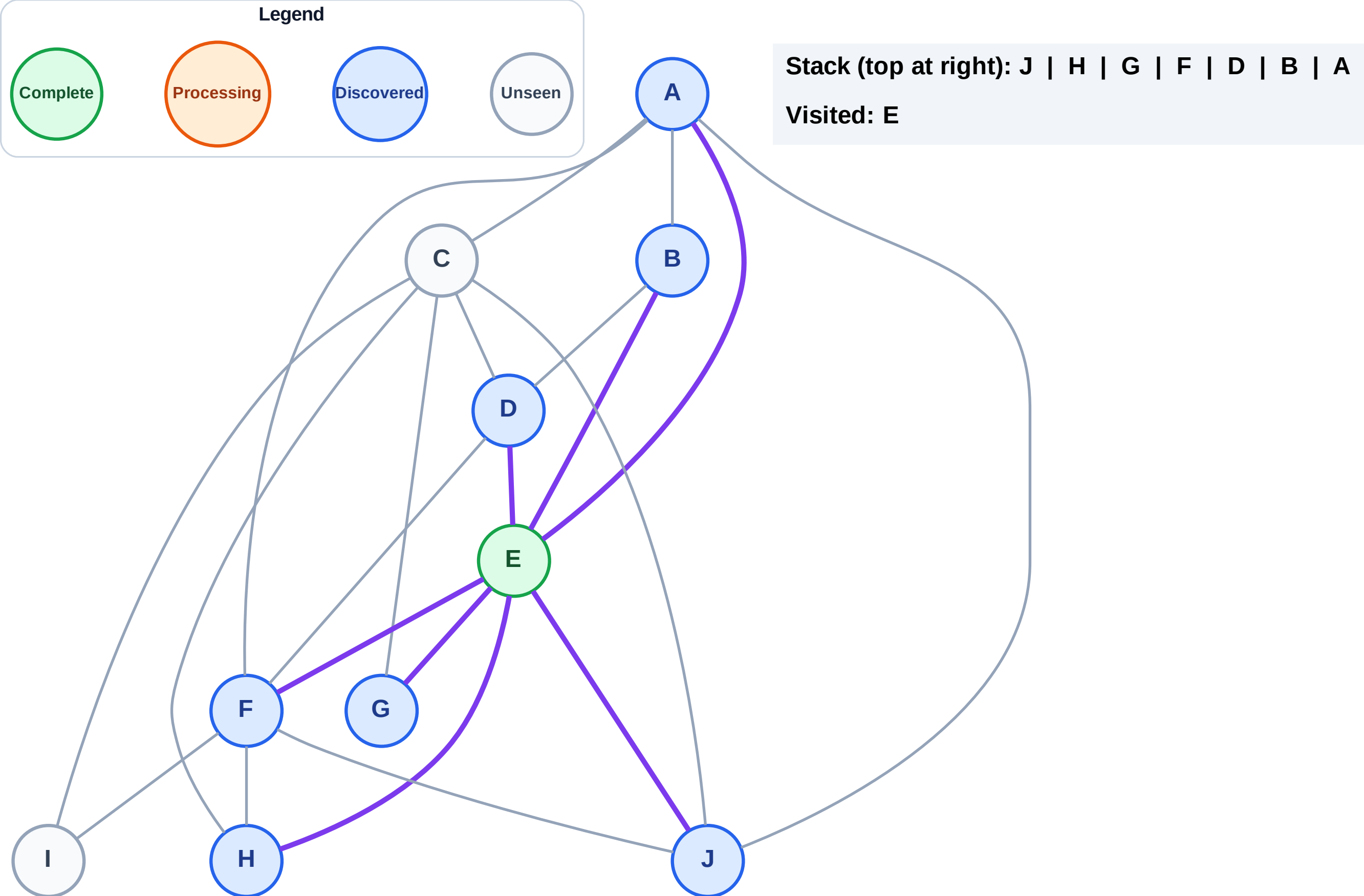
Stack (top at right): (empty)

Visited: (none)



# DFS Traversal

### Step 3: Discover J, H, G, F, D, B, A from E



# DFS Traversal

Step 4: Process node A

Legend

Complete

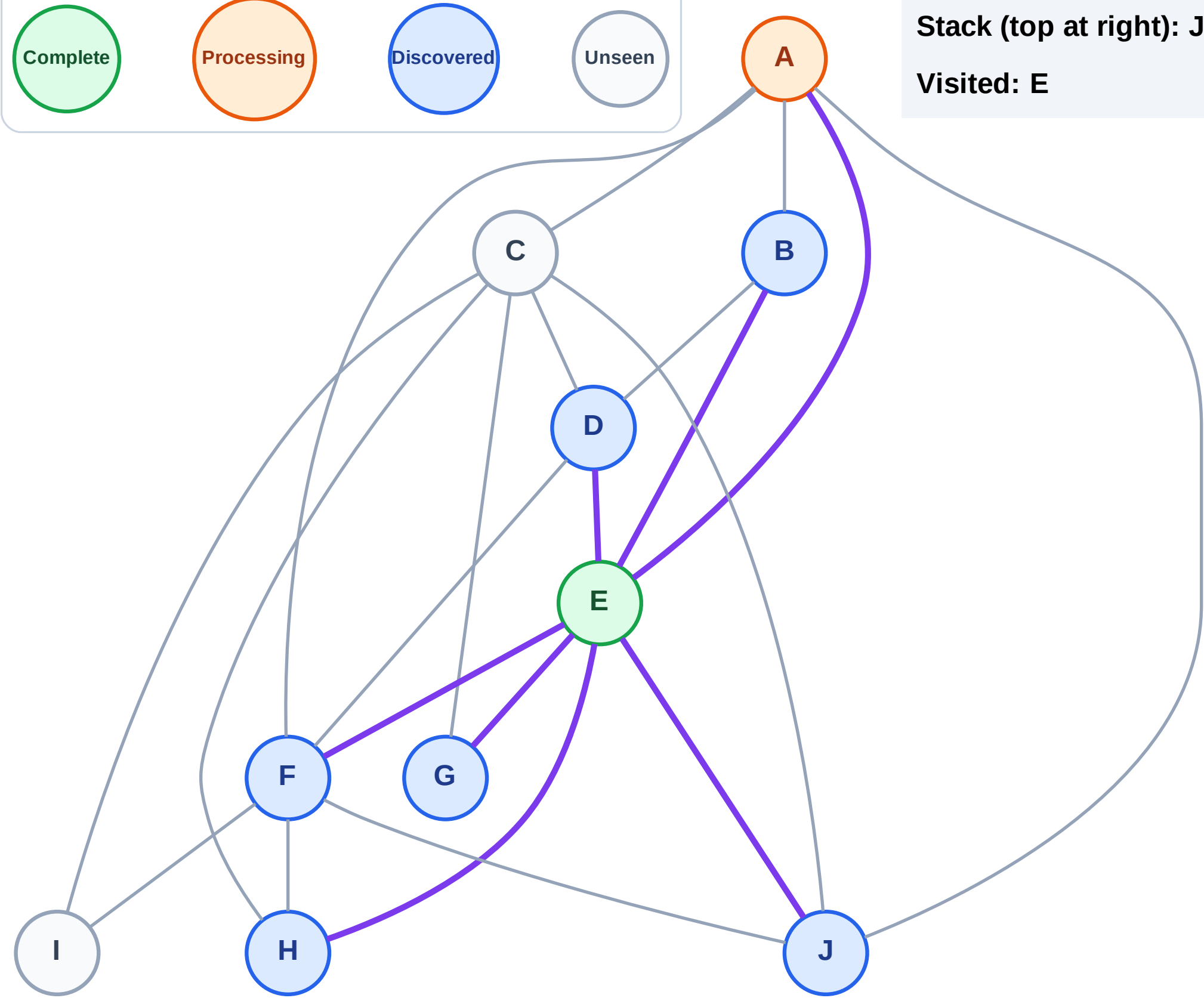
Processing

Discovered

Unseen

Stack (top at right): J | H | G | F | D | B

Visited: E



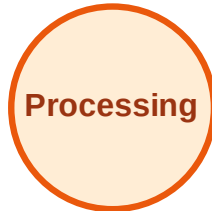
# DFS Traversal

Step 5: Discover C from A

## Legend



Complete



Processing



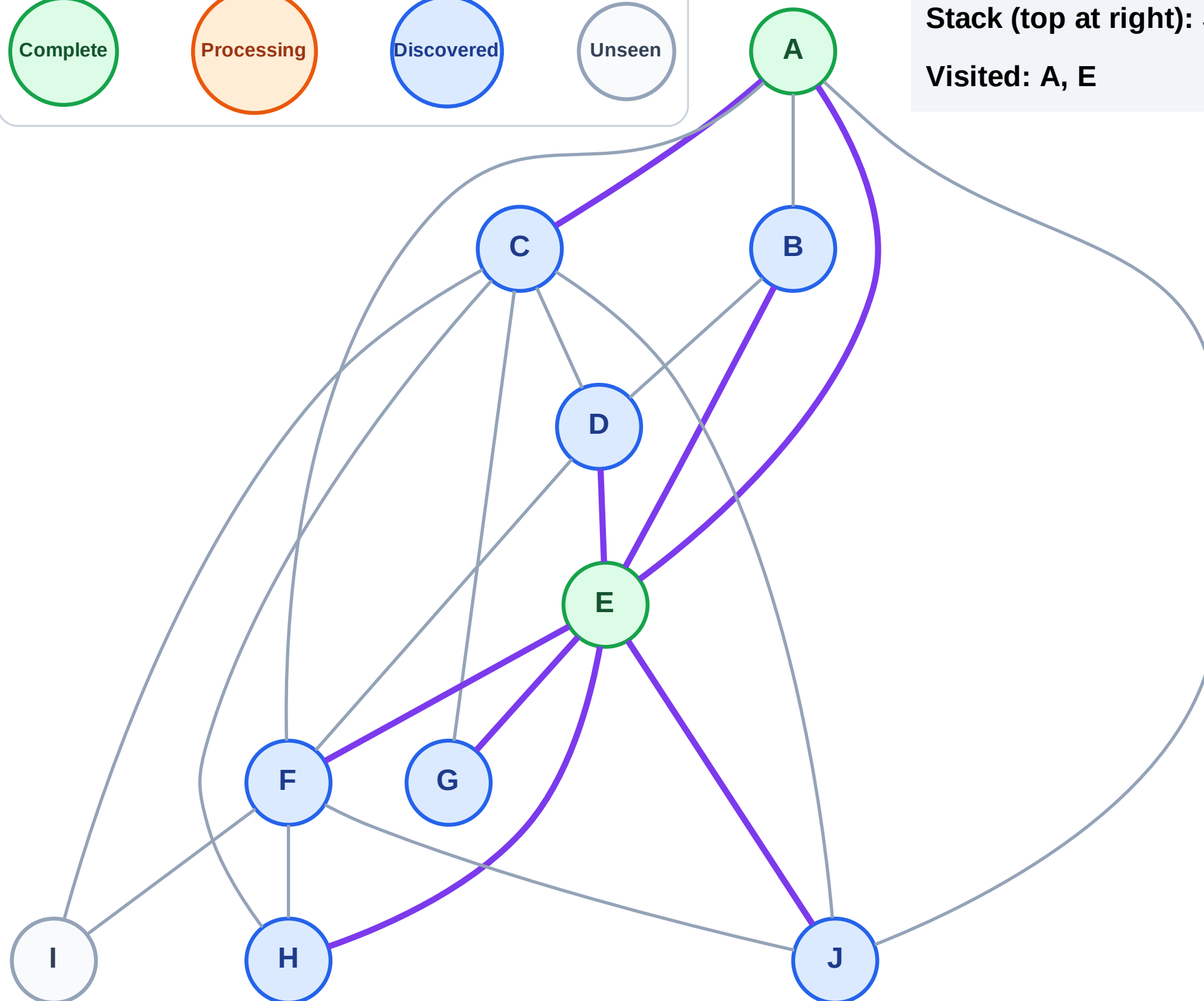
Discovered



Unseen

Stack (top at right): J | H | G | F | D | B | C

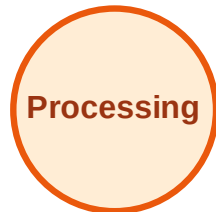
Visited: A, E



# DFS Traversal

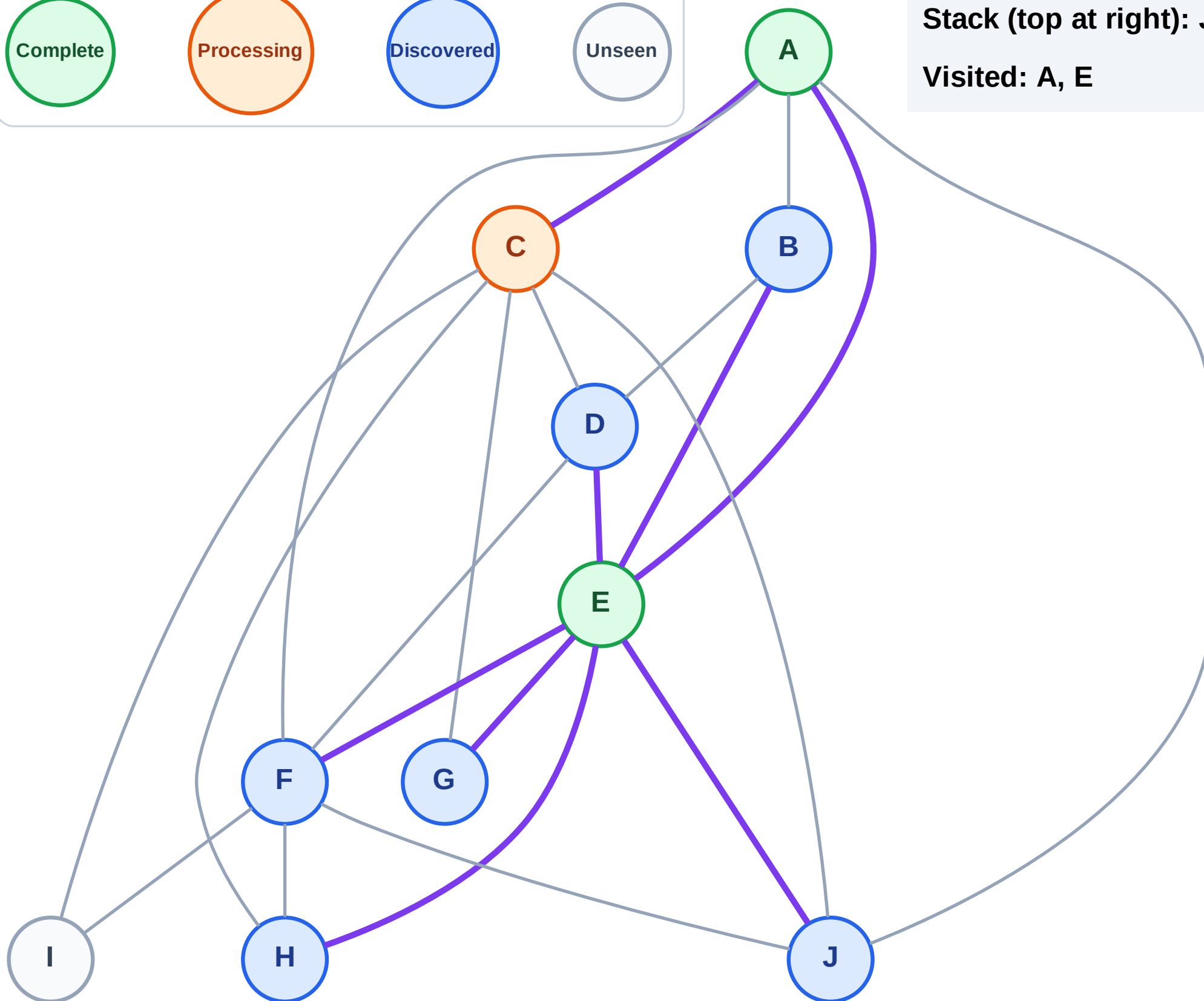
Step 6: Process node C

## Legend



Stack (top at right): J | H | G | F | D | B

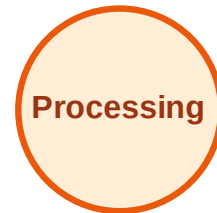
Visited: A, E



# DFS Traversal

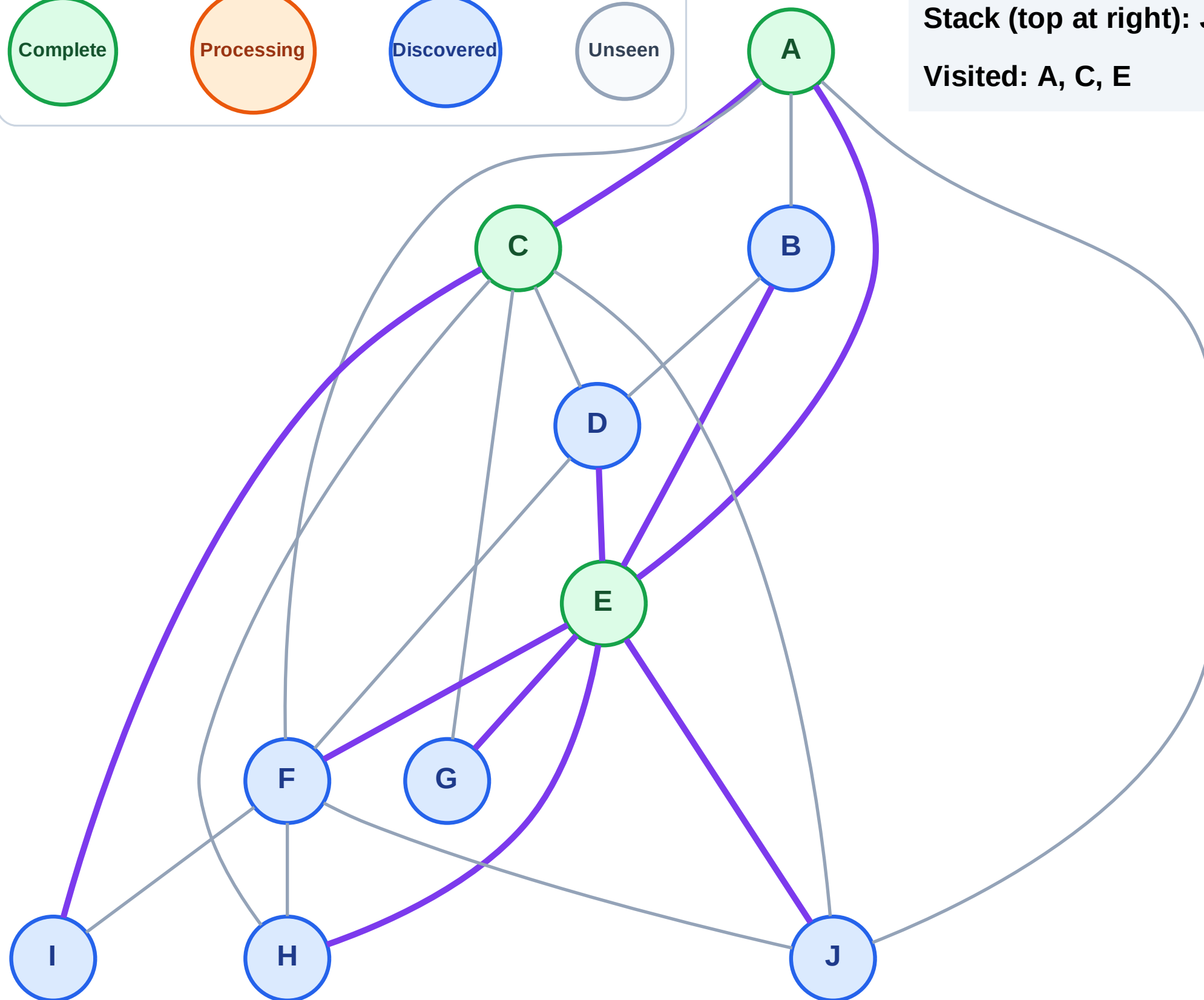
Step 7: Discover I from C

## Legend



Stack (top at right): J | H | G | F | D | B | I

Visited: A, C, E



# DFS Traversal

Step 8: Process node I

Legend

Complete

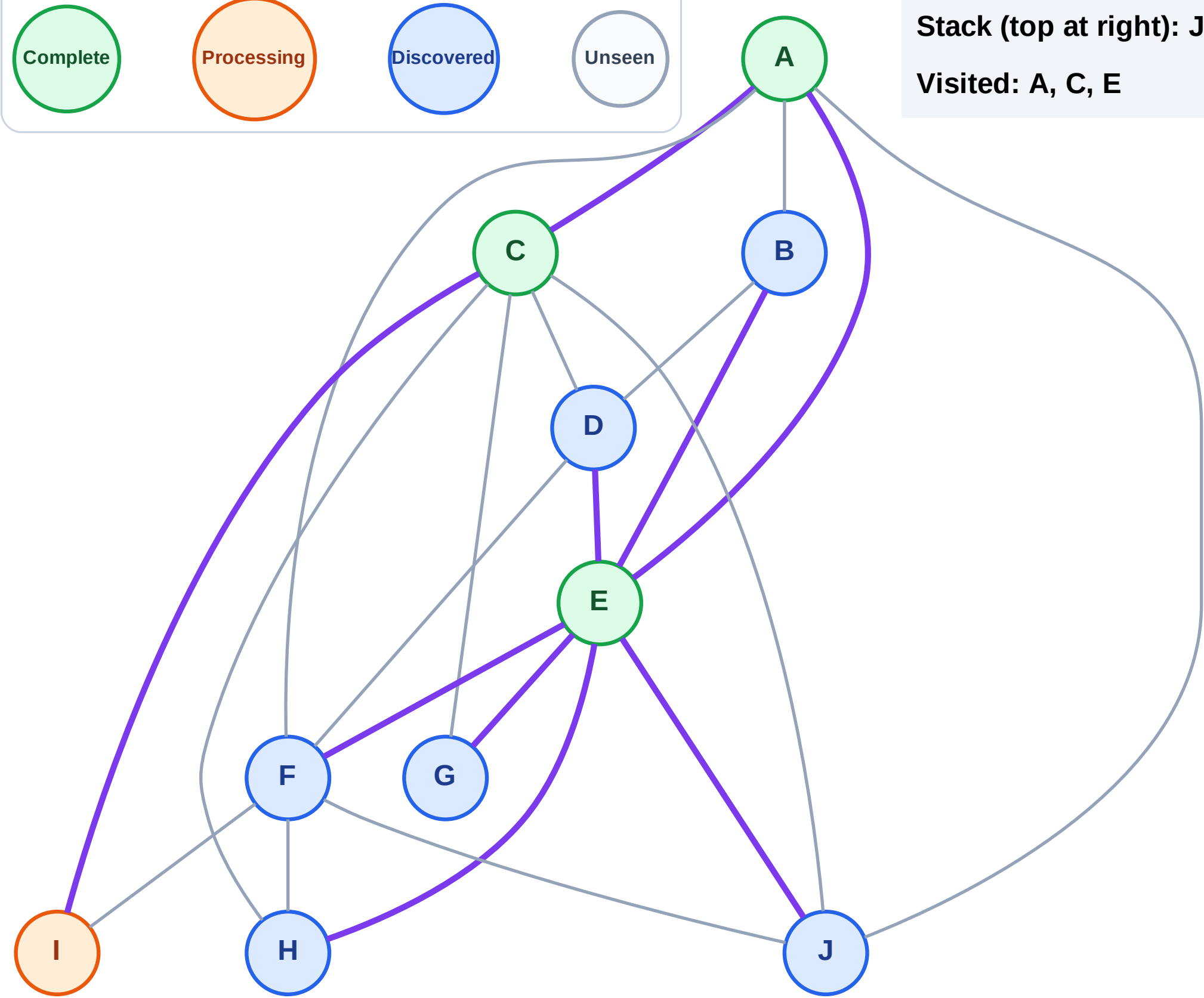
Processing

Discovered

Unseen

Stack (top at right): J | H | G | F | D | B

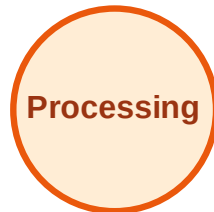
Visited: A, C, E



# DFS Traversal

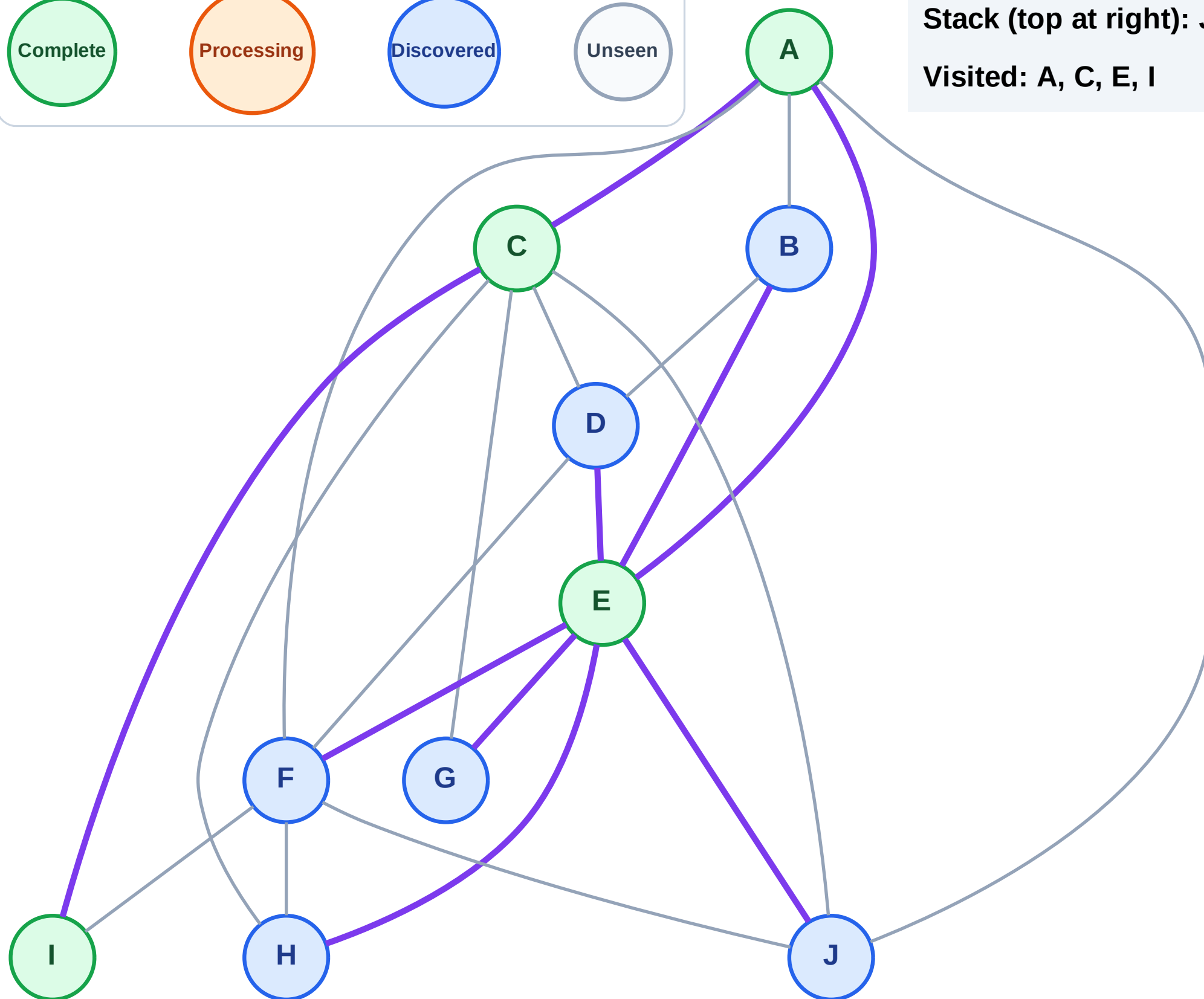
Step 9: No new neighbors from I

## Legend



Stack (top at right): J | H | G | F | D | B

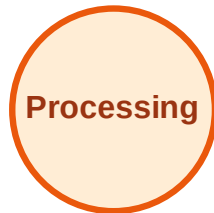
Visited: A, C, E, I



# DFS Traversal

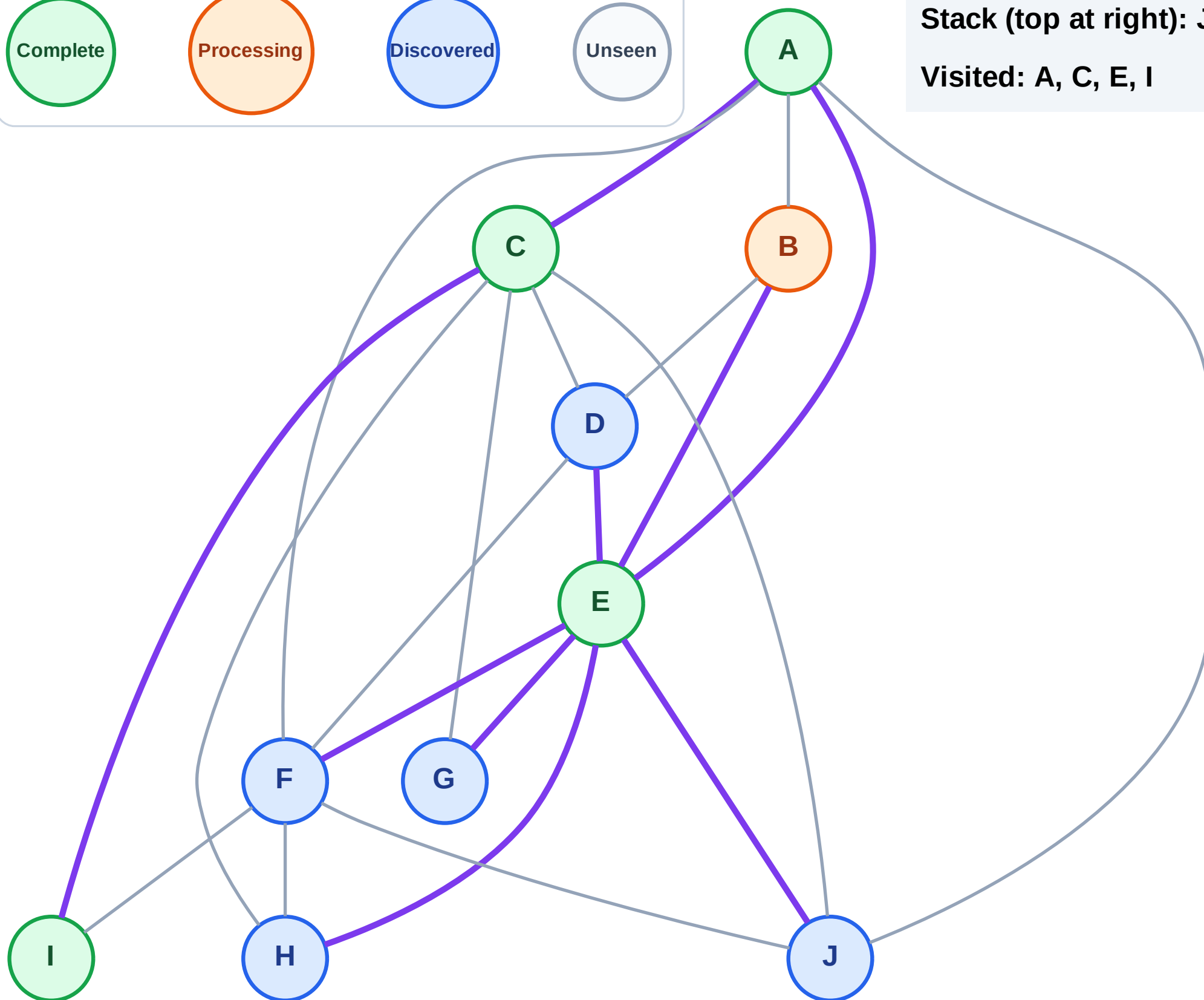
Step 10: Process node B

## Legend



Stack (top at right): J | H | G | F | D

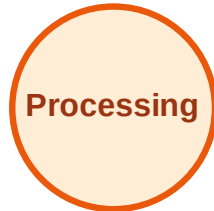
Visited: A, C, E, I



# DFS Traversal

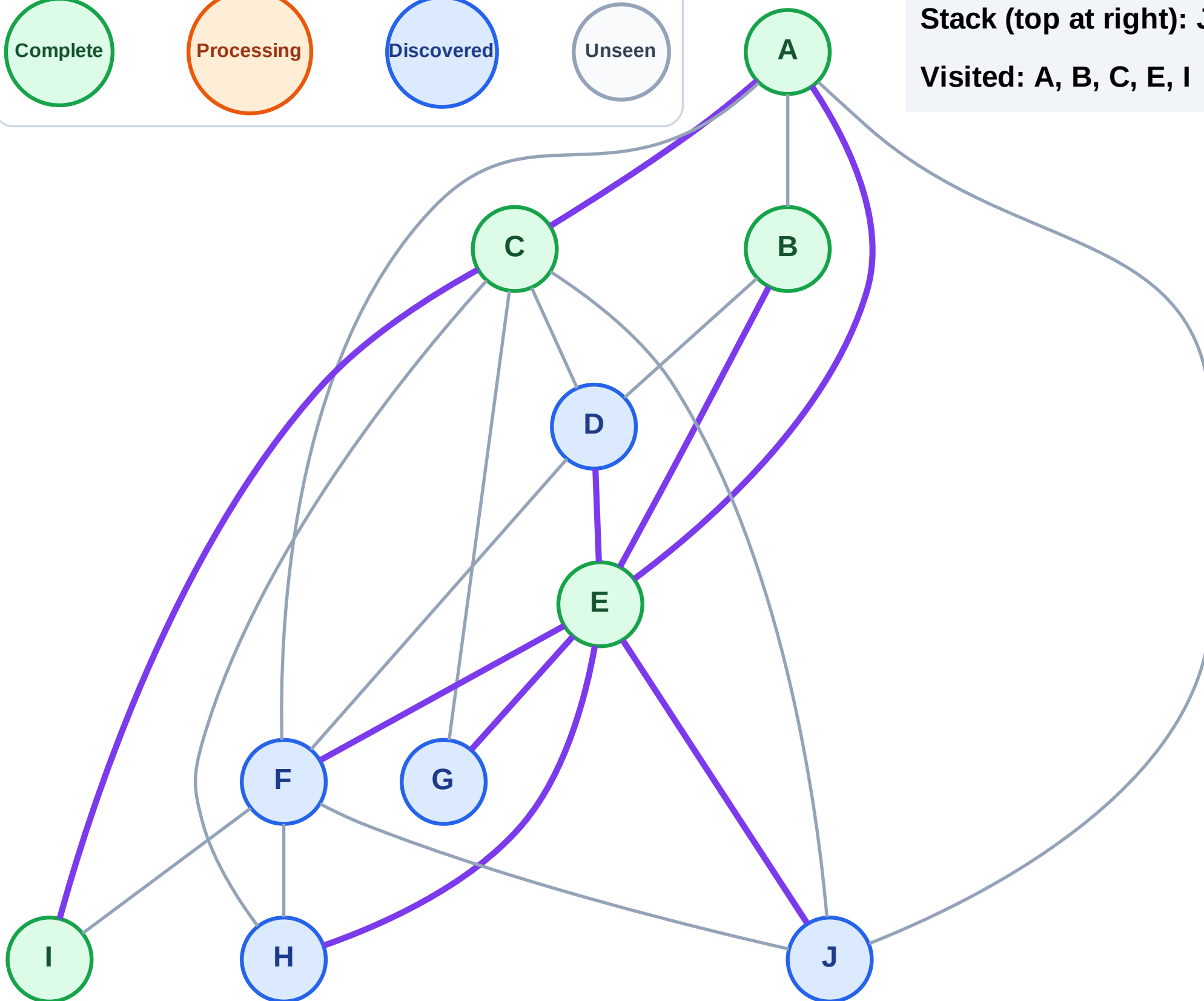
Step 11: No new neighbors from B

## Legend



Stack (top at right): J | H | G | F | D

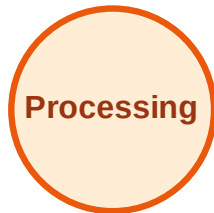
Visited: A, B, C, E, I



# DFS Traversal

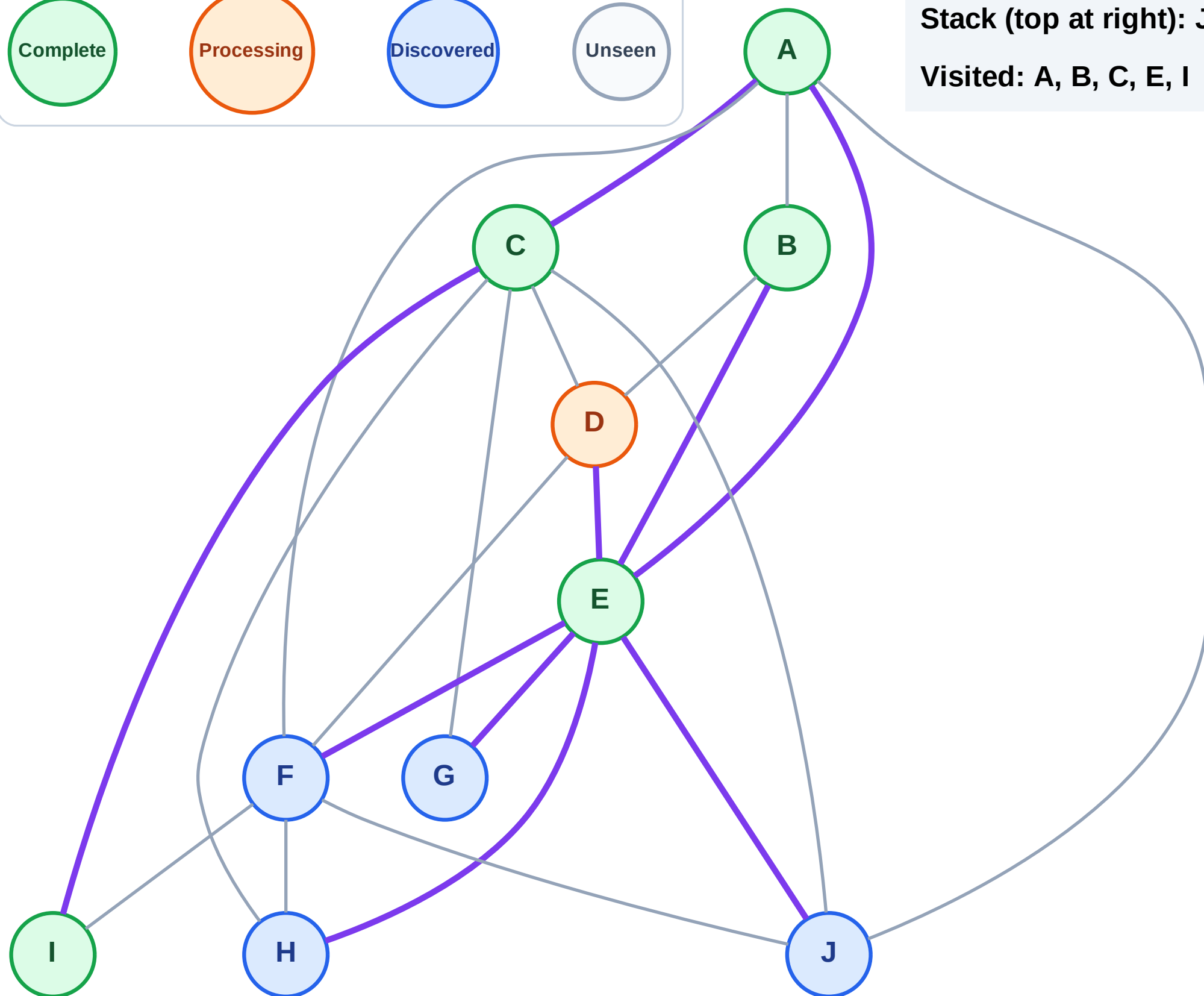
Step 12: Process node D

## Legend



Stack (top at right): J | H | G | F

Visited: A, B, C, E, I



# DFS Traversal

Step 13: No new neighbors from D

Legend

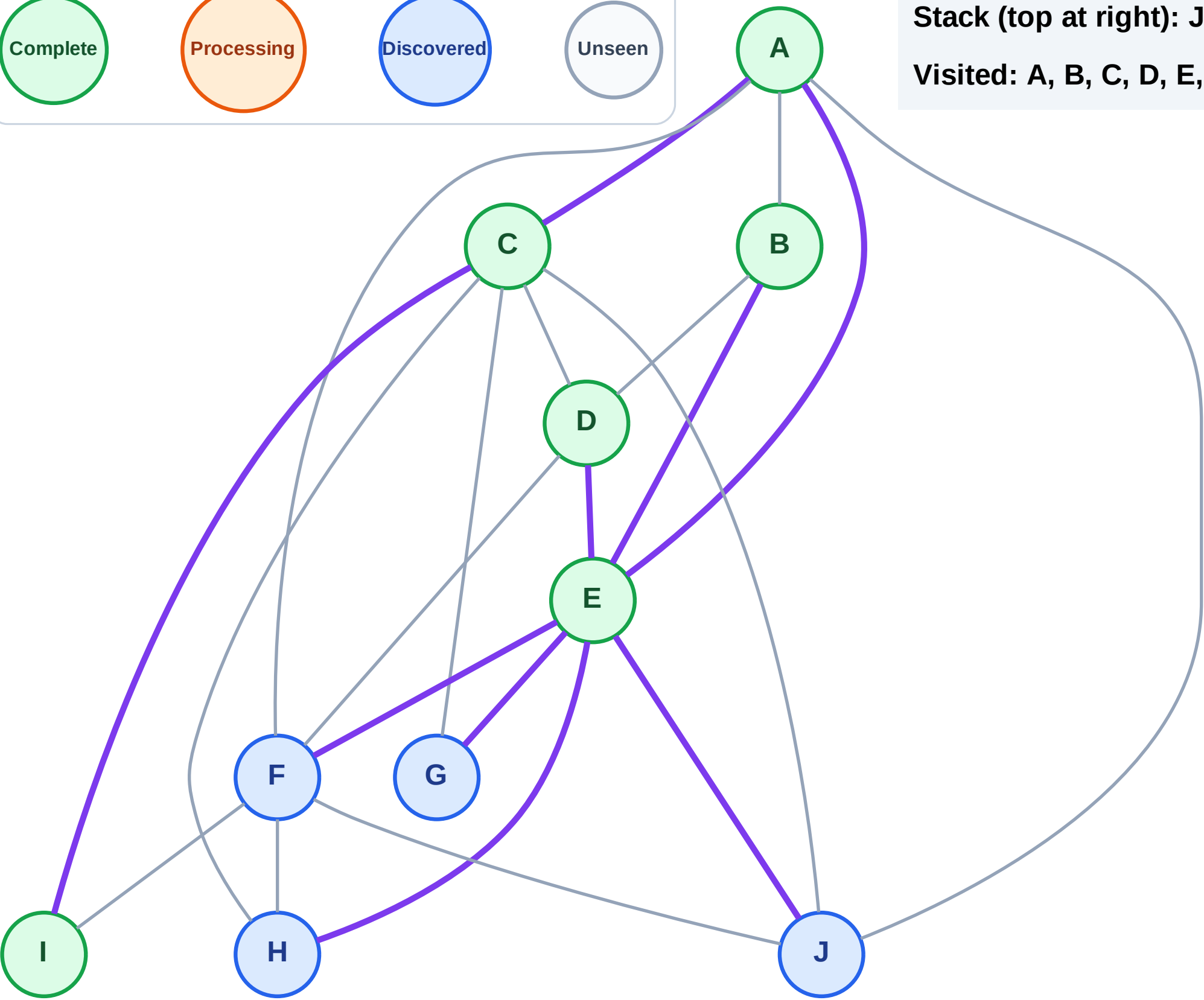
Complete

Processing

Discovered

Unseen

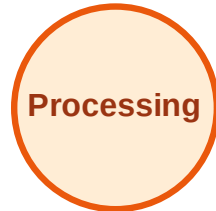
Stack (top at right): J | H | G | F  
Visited: A, B, C, D, E, I



# DFS Traversal

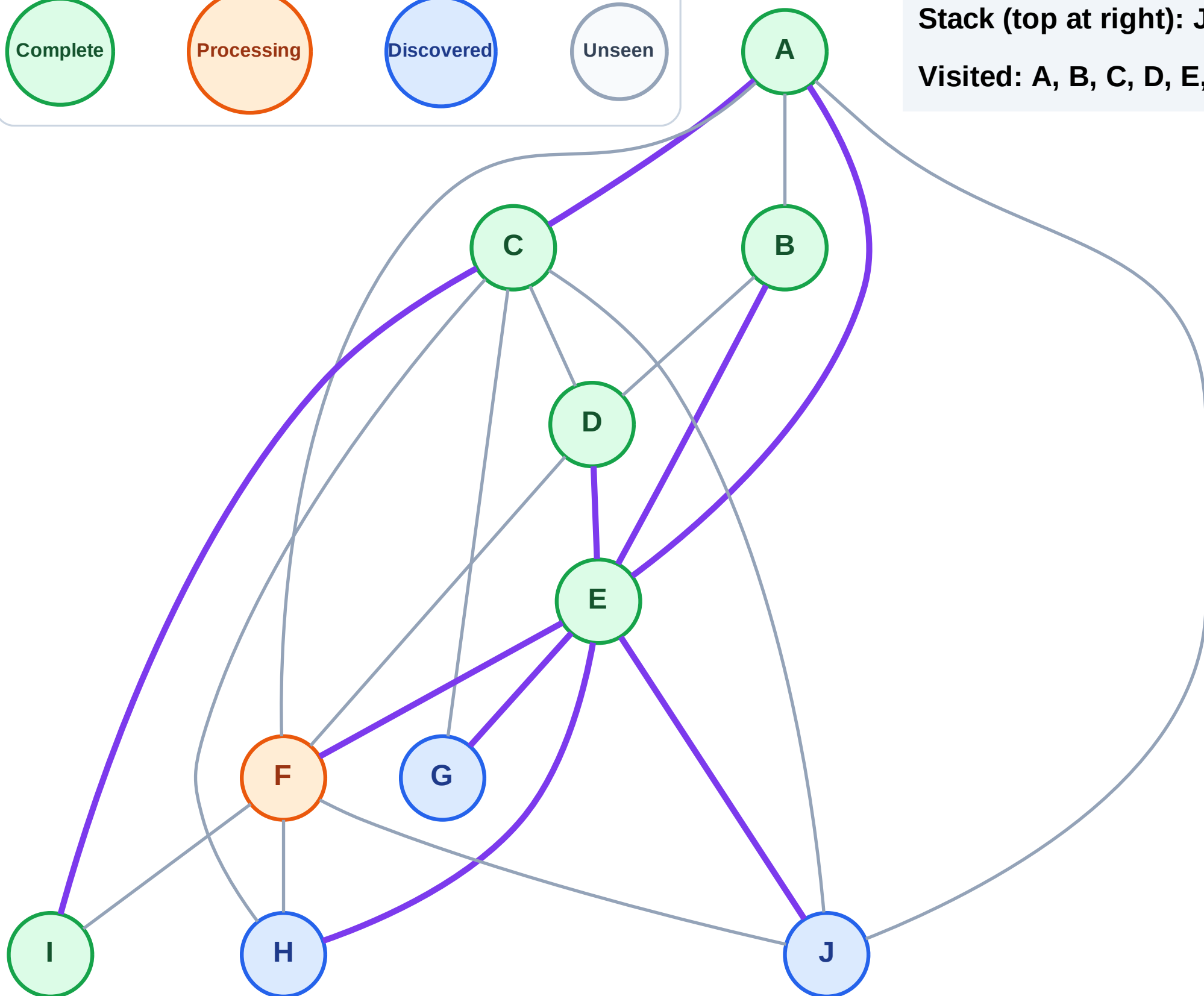
Step 14: Process node F

## Legend



Stack (top at right): J | H | G

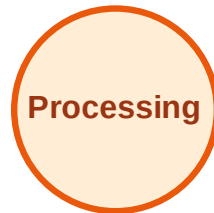
Visited: A, B, C, D, E, I



# DFS Traversal

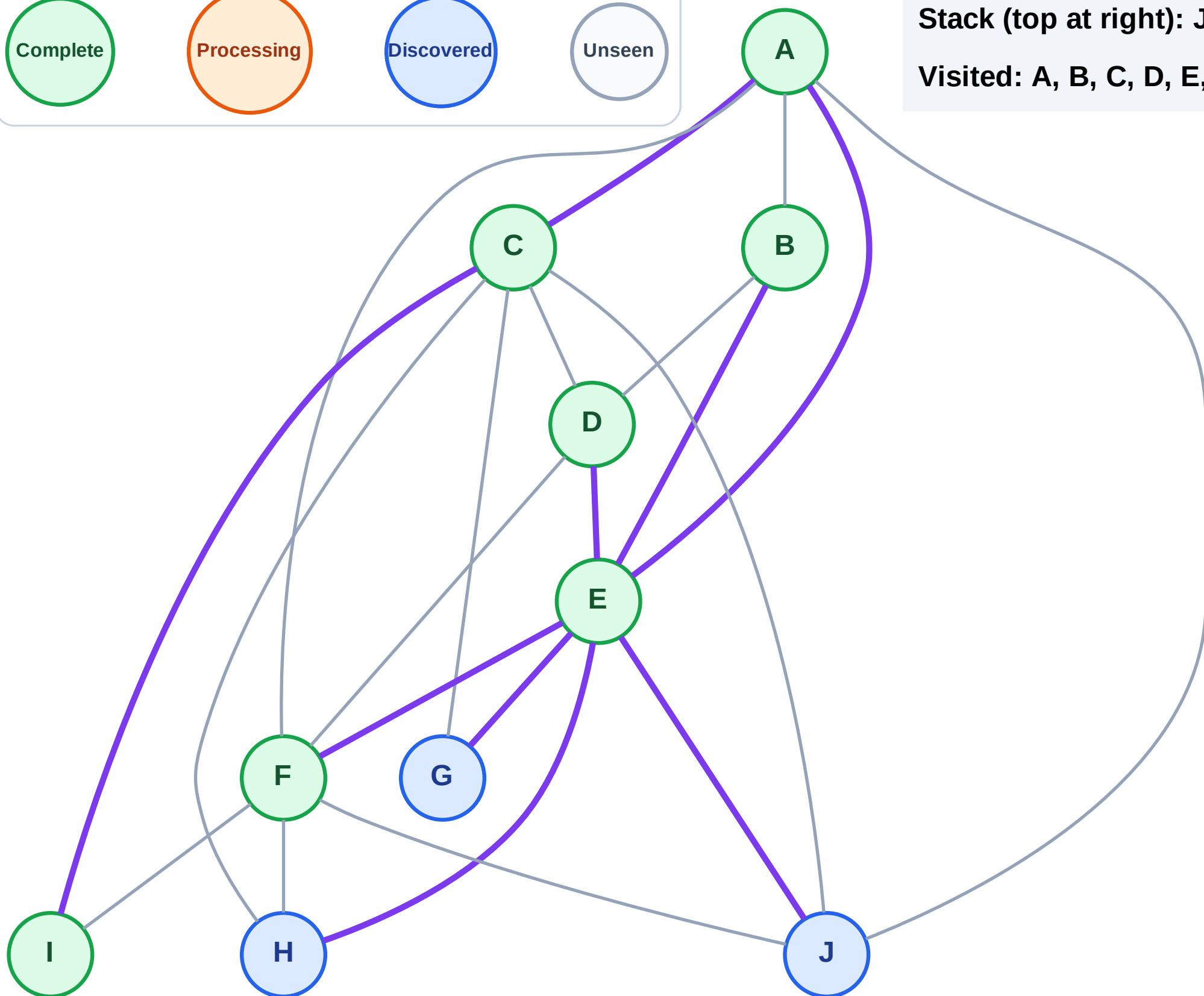
Step 15: No new neighbors from F

## Legend



Stack (top at right): J | H | G

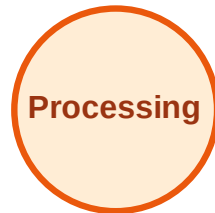
Visited: A, B, C, D, E, F, I



# DFS Traversal

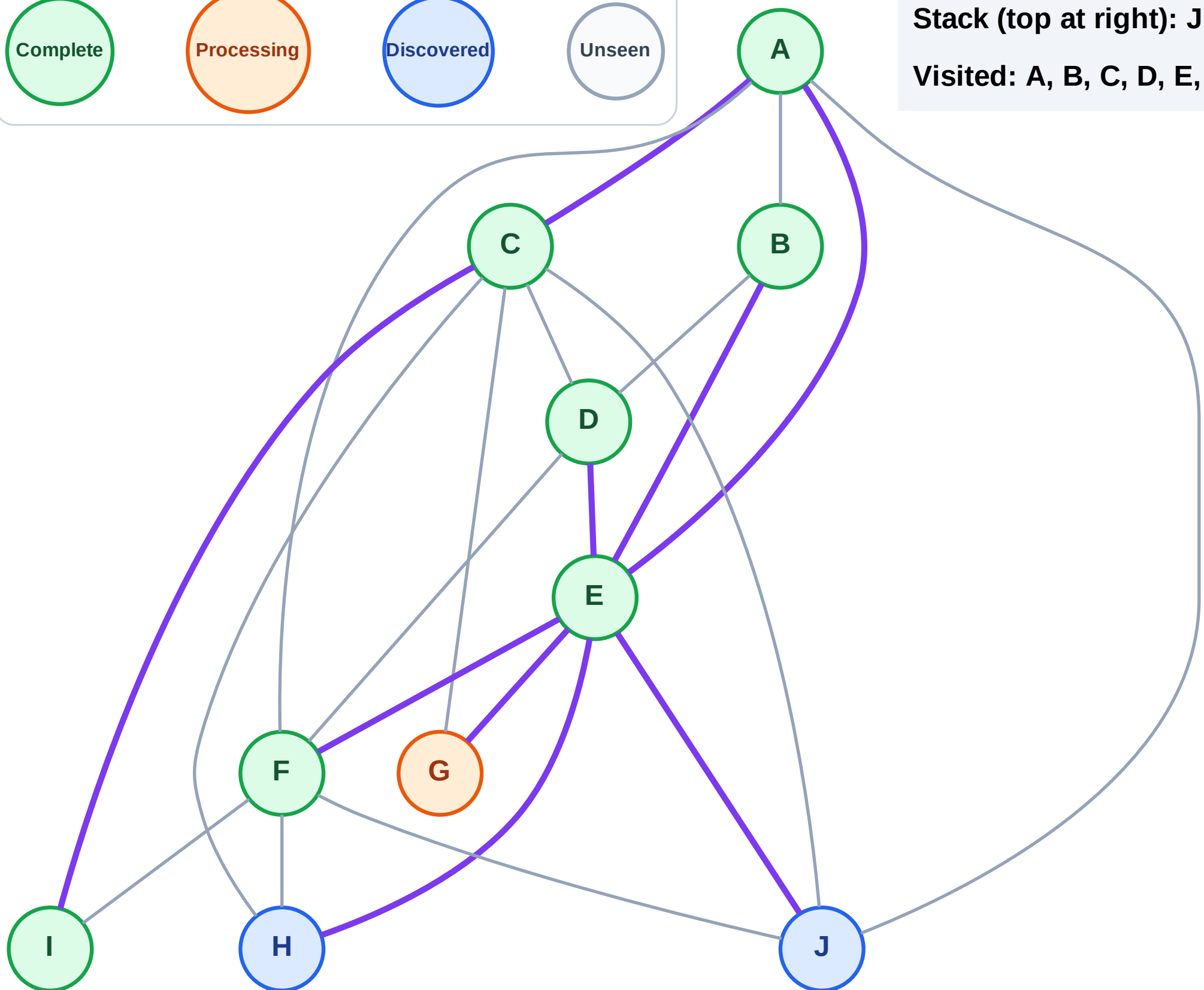
Step 16: Process node G

## Legend



Stack (top at right): J | H

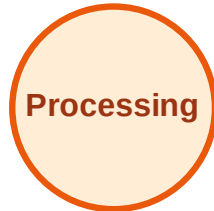
Visited: A, B, C, D, E, F, I



# DFS Traversal

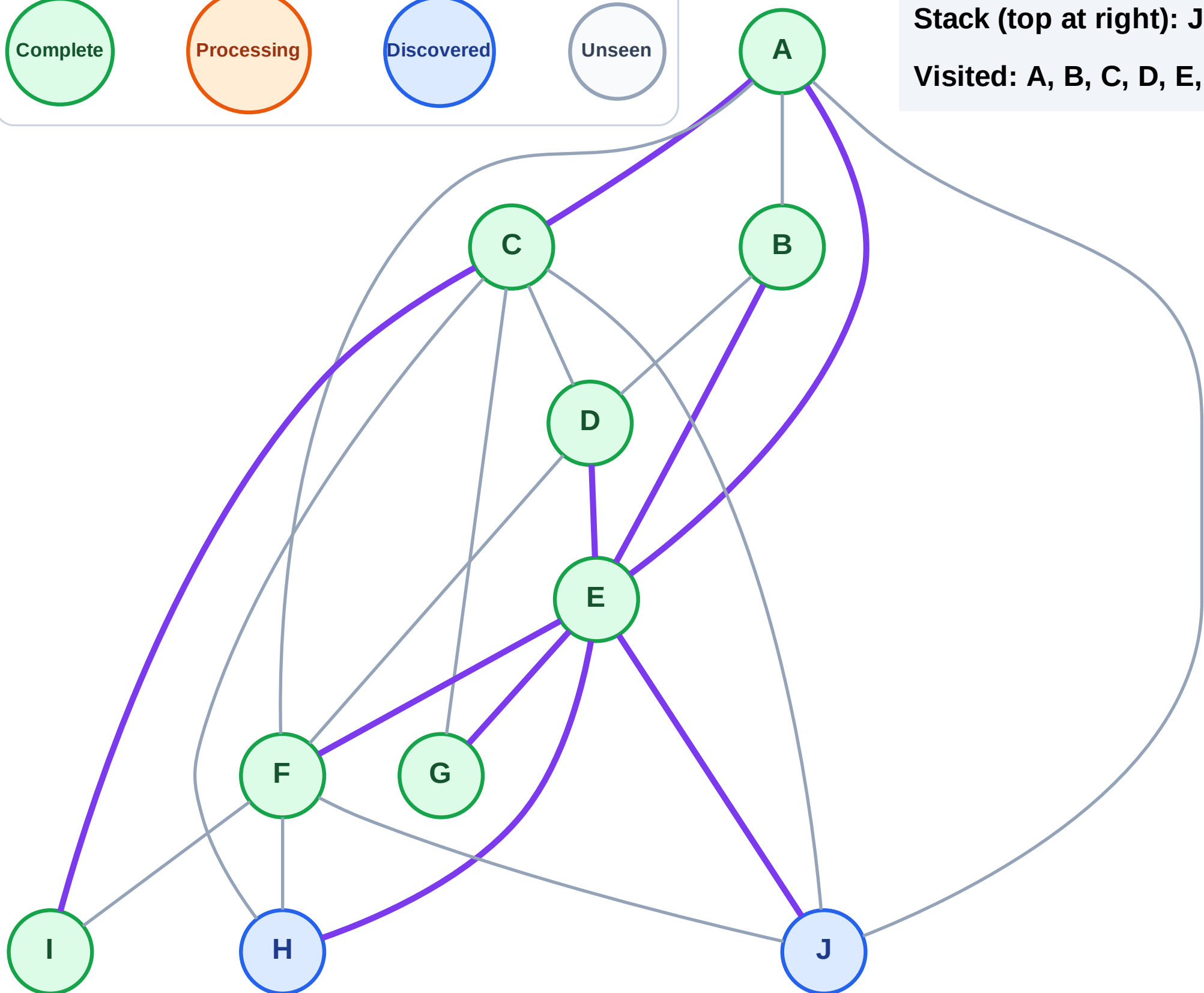
Step 17: No new neighbors from G

## Legend



Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



# DFS Traversal

Step 18: Process node H

Legend

Complete

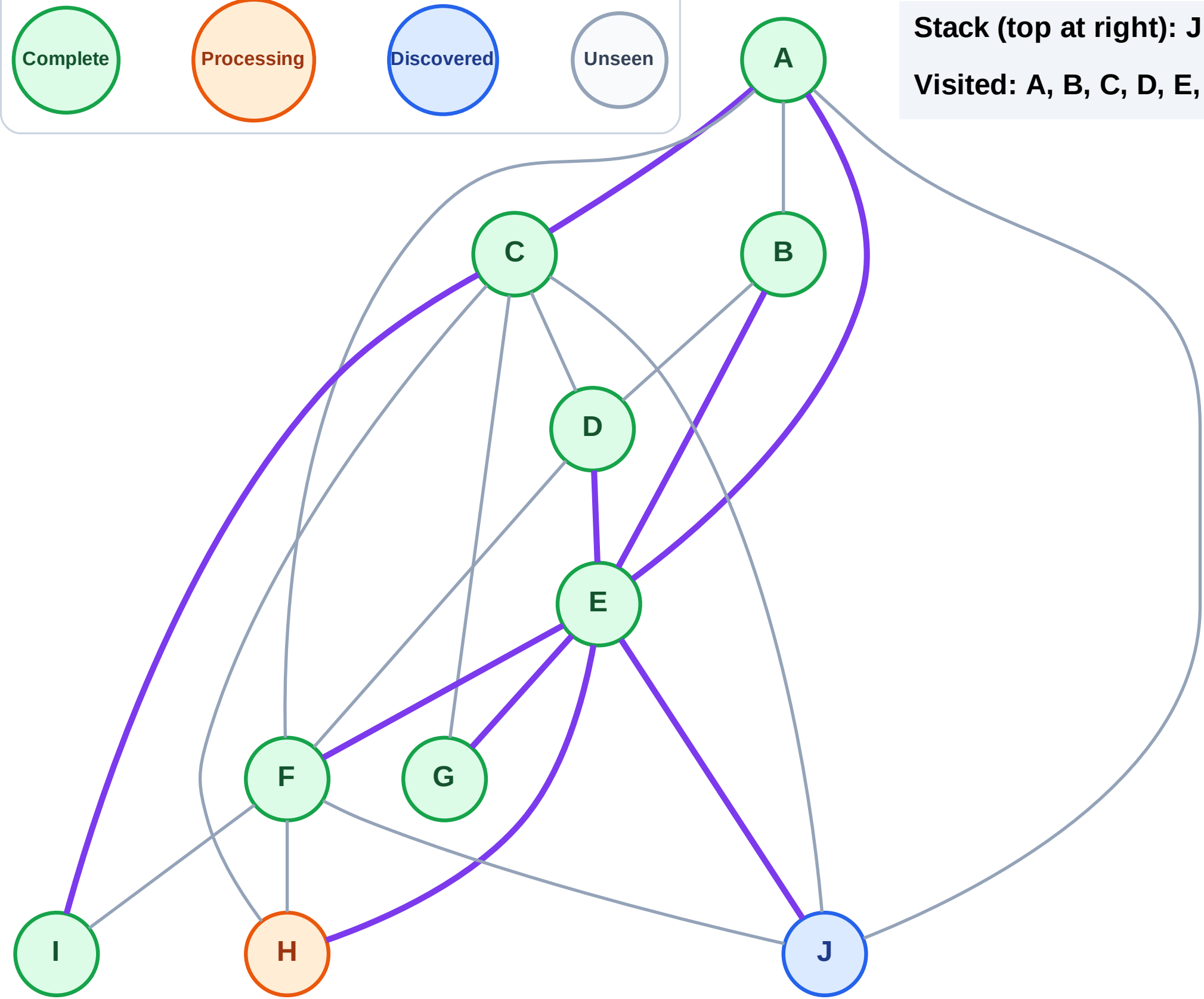
Processing

Discovered

Unseen

Stack (top at right): J

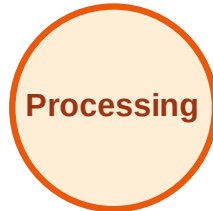
Visited: A, B, C, D, E, F, G, I



# DFS Traversal

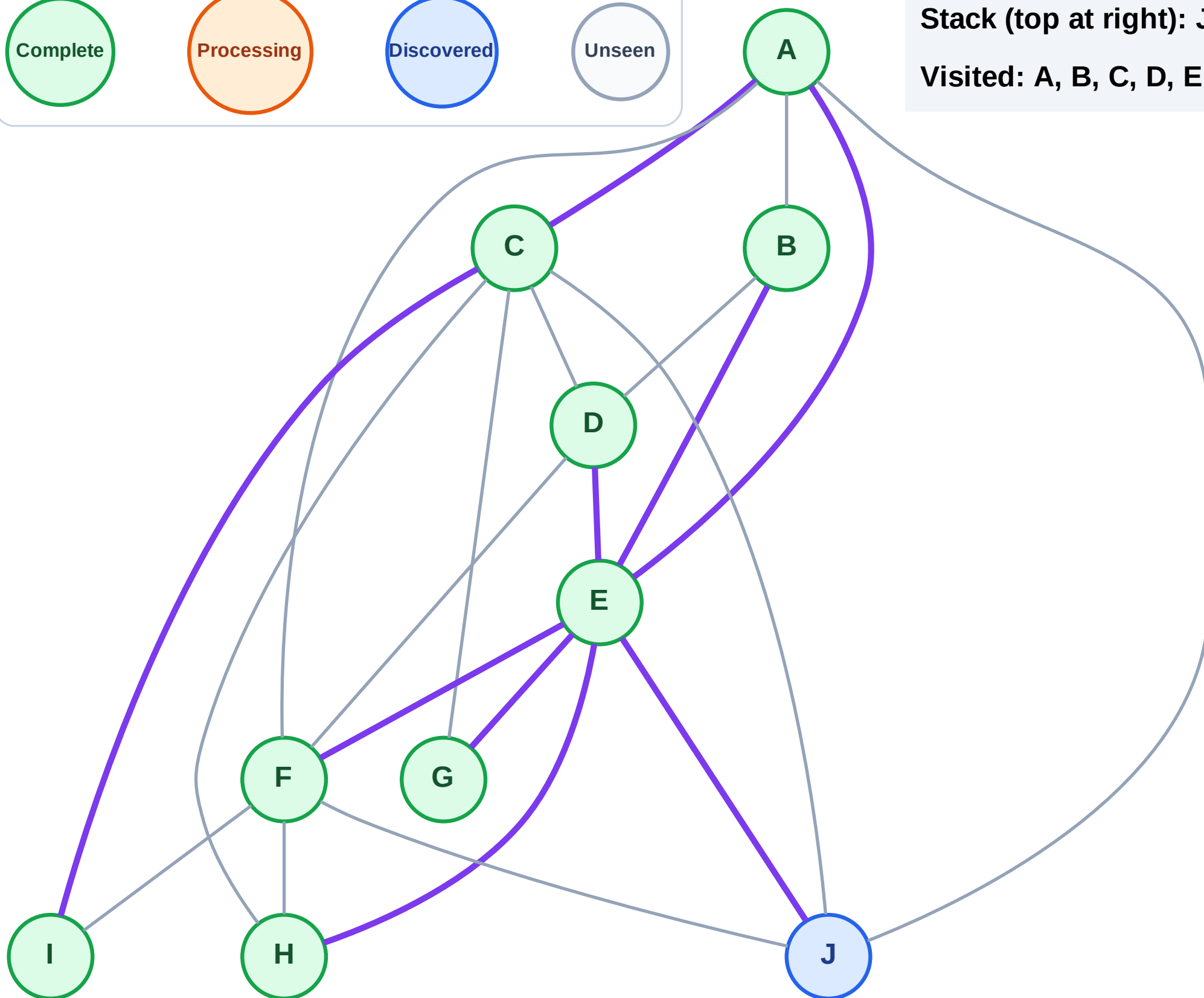
Step 19: No new neighbors from H

## Legend



Stack (top at right): J

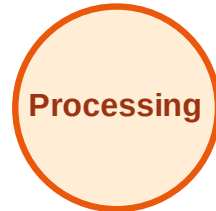
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

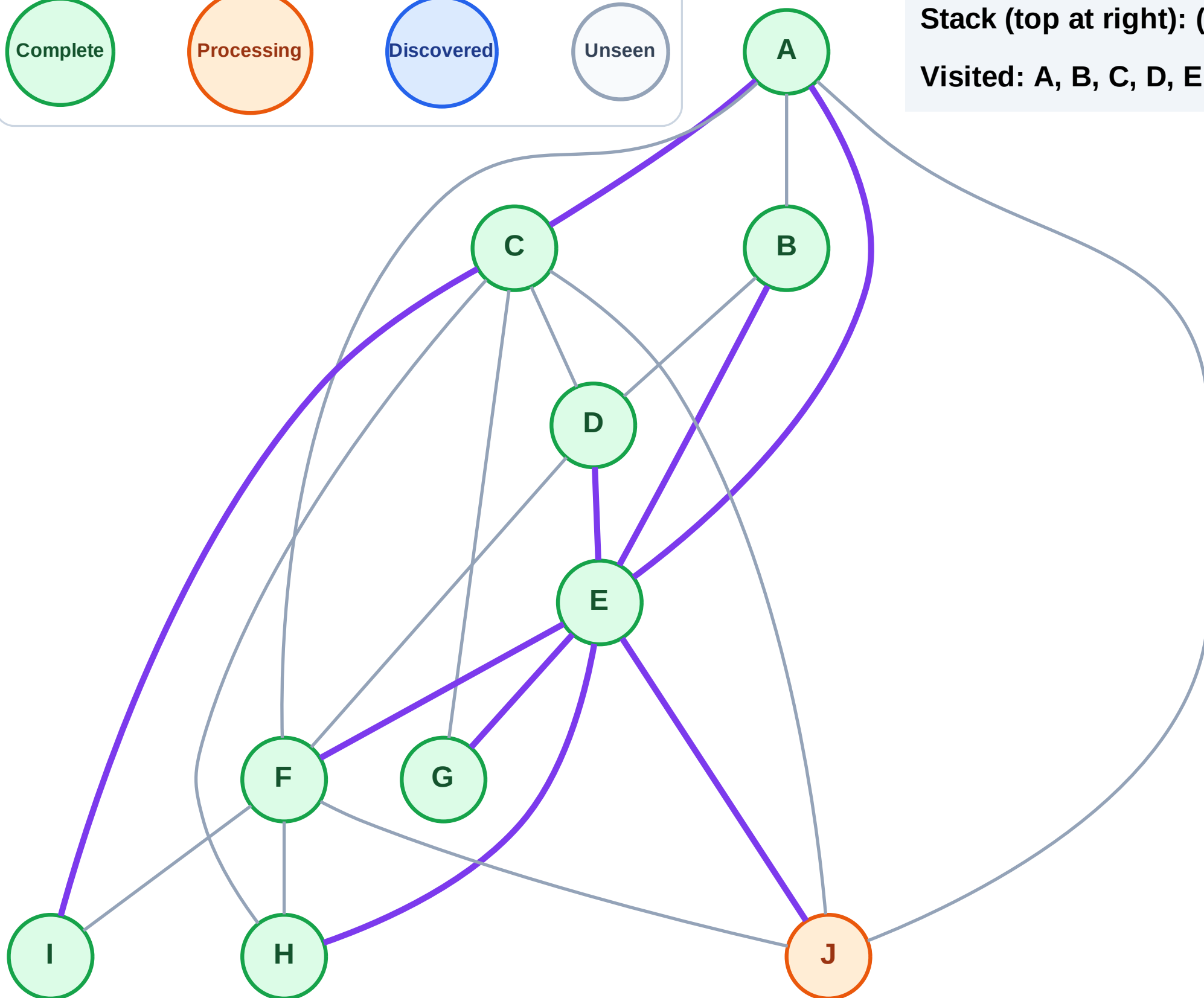
Step 20: Process node J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

